Exploring equity standards for transport policy: A case study of objective accessibility, perceived accessibility, and employment outcomes in the Greater Toronto and Hamilton Area, Canada

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Abstract

The lack of standards for the equitable distribution of accessibility complicates efforts to address transport poverty. This study presents a model-based approach for selecting equity standards. This is done by exploring the relationship between objective accessibility and two employment-related variables for a sample of residents in the Greater Toronto and Hamilton Area, Canada. We use objective accessibility calculated for three modes of travel (active transportation, public transit, and private vehicles) and other individual-level variables to estimate ordinal models. Further, we use spatial filters as instrumental variables to account for potential endogeneity. We examine the association between objective accessibility and the perception of having too few jobs available within an ideal commuting time (a perceived accessibility indicator) and the reports of having declined jobs because of transportation constraints (an employment outcome indicator). The results indicate that higher transit accessibility is associated with more positive perceived accessibility, whereas associations with active accessibility depend on whether respondents travel actively. Based on the results of these models, we identify public transit accessibility thresholds that reduce the likelihood of negative perceptions for transport-disadvantaged groups. We propose three thresholds that we classify as conservative, intermediate, and progressive. These thresholds are an example of a model-driven approach to support transportation planning in achieving more equitable outcomes, as well as for participatory processes to collectively define minimum or sufficient accessibility levels.

Keywords: benchmark, justice, job, egalitarian, sufficient

1. Introduction

Transportation equity refers to the distribution of the burdens and benefits of transportation systems among a population (Karner et al., 2020; Pereira et al., 2017). Burdens include health-related concerns arising from exposure to air pollution, noise, and traffic; sedentary lifestyles; and the risk of injury or death from collisions. In contrast, the primary benefit is the provision of accessibility (Pereira and Karner, 2021), understood as the ease of reaching (spatially distributed) opportunities and destinations (Páez et al., 2012; Hansen, 1959; Soukhov et al., 2025a), including employment, healthcare, educational services, and venues for social interaction.

Accessibility is one of the primary outcomes of spatial development (Páez et al., 2012). It is influenced by the spatial distribution of the population and opportunities and depends on the characteristics and performance of the transportation network. In other words, it is a product of both transportation systems and land use patterns (Wee and Geurs, 2011). There is a growing consensus among researchers

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and transportation planners that the goal of transport policy should be to improve accessibility (El-Geneidy and Levinson, 2022; Boisjoly and El-Geneidy, 2017), since activity participation is one of the main ways in which transportation can help to promote a dignified life (Karner et al., 2020; Dos Santos et al.). In contrast, insufficient or inadequate accessibility can prevent individuals from engaging in activities that are meaningful to them, for reasons that are beyond their control. This process is described in the literature as Transport-Related Social Exclusion (TRSE) (Litman, 2024; Lucas, 2012; Preston and Rajé, 2007; Luz and Portugal, 2022; Paez et al., 2009). Insufficient accessibility, when combined with other forms of social and economic disadvantage, can result in transportation poverty (Lucas, 2019; Luz and Portugal, 2022): the failure of transportation and land use systems to provide access to essential services, employment, and other activities. Transportation poverty increases vulnerability among individuals who already face challenges associated with age (van Dülmen et al., 2022), income (Allen and Farber, 2019), race (Awaworyi Churchill, 2020), gender (Murphy et al., 2023), disability (Upham et al., 2022), among other factors (Kim et al., 2024). With the suburbanization of low-income populations, it is estimated that over one million Canadians live in transport poverty (Allen and Farber, 2019).

A challenge with the use of accessibility as a measure of transportation poverty is the lack of standards for the equitable distribution of accessibility (Soukhov et al., 2025b). Additionally, there is an absence of transportation data in the form of open data products to support urban planners (Arribas-Bel et al., 2021; Behrendt and Sheller, 2024; Páez, 2021), and transportation planners receive limited formal guidance in assessing the equity implications of their projects (Miller, 2018; Palm et al., 2023; te Brömmelstroet et al., 2016). Furthermore, relatively few studies explicitly link accessibility levels with the experience of accessibility, such as employment opportunities declined or accepted due to transport conditions, or the ease to reach employment opportunities within ideal commuting times. Combined, this results in a gap in the development and implementation of accessibility standards to ensure equity in transportation planning and policy.

This study is guided by the following research questions: How do objective job accessibility indicators, i.e., calculated based on spatial and transport data, relate to perceived accessibility and to self-reported employment outcomes? And could a given level of accessibility serve as a standard or goal for equitable transport planning? This study proposes a model-based methodology for defining equity standards in transportation planning. We obtain these standards from the estimated relationships between objective job accessibility, people's perceived accessibility, and a self-reported employment outcome.

While prior studies have examined the association between objective and perceived accessibility (Pot et al., 2021; Lättman et al., 2016; Zhu and Lucas, 2025), and between objective accessibility and employment outcomes (Bastiaanssen et al., 2021; Duarte et al., 2023; Jin and Paulsen, 2018; Johnson et al., 2017; Matas et al., 2010; Kain, 1968; Bania et al., 2008; Delmelle et al., 2021; Merlin and Hu, 2017; Bastiaanssen et al., 2025), few have used these relationships to derive accessibility thresholds (e.g., Negm and El-Geneidy (2026)). Grounded in a transport equity perspective, the novelty of our work is the use of a model-based approach to define accessibility thresholds conditioned on individual characteristics, allowing standards to be defined from the standpoint of a chosen reference profile. Specifically, we define accessibility thresholds as the minimum levels at which the reference profile becomes more likely to give a non-negative response than a negative one, or, under a more demanding definition, a positive response than a negative one. Although these thresholds are obtained at the neighbourhood level, they indirectly incorporate individual-level characteristics, since they are derived from individuals' self-reported perceptions and outcomes.

This study is guided by an understanding of accessibility as a means through which transportation can foster human dignity (Pereira and Karner, 2021; Dos Santos et al.), combined with an egalitarian and sufficientarian view of transportation equity. From this perspective, a fair transportation system should guarantee a minimum, sufficient level of accessibility to all members of society, with this sufficient level defined from the standpoint of those historically and currently facing the greatest transportation disadvantage (Martens and Golub, 2021).

After this introduction, we structured this article as follows: Section 2 provides a brief theoretical background on objective and perceived accessibility, on the identified connections between objective job accessibility and employment outcomes, and on the identification of equity standards based on accessibility. Section 3 presents the materials and methods, covering the study area and data sources, the method used to

calculate accessibility, how we modelled the relationship between objective accessibility and the dependent variables, the process used to identify equity standards, and how we measured transportation equity based on these standards. Section 4 presents the results. Section 5 critically analyzes the paper and presents its limitations. Finally, Section 6 concludes with final considerations.

2. Theoretical background

2.1. Objective and perceived accessibility

The most commonly used accessibility measures are based on the gravity model of Hansen (1959). This model considers the number of opportunities that a person can reach to be proportional to the sum of opportunities in each destination, discounted by the travel cost to reach that destination. This type of metric is classified as a place-based measure (Pereira and Herszenhut, 2023), representing a characteristic of a given location that assumes that all people in this location have equal access to the spatially distributed opportunities. For relying on the use of spatial and transport data to create indicators, these measures are referred to as calculated, spatial, location-based, or objective accessibility; hereafter, we use the term *objective accessibility*.

However, objective accessibility does not take into account the individual characteristics of people. The perceived accessibility, on the other hand, refers to the perception of people of how easily they can reach opportunities (Lättman et al., 2016; Pot, 2025; Mehdizadeh et al., 2025), resulting from not only the interaction of land-use and transport infrastructure (Geurs and van Wee, 2004), but also from travel options (e.g., having a car or a transit pass), travel experiences, and individual characteristics and preferences. Often, perceived accessibility deviates from objective accessibility, with some research indicating that objective accessibility might not be able to capture the multiple ways in which people experience a lack of access to opportunities (Zhu and Lucas, 2025). Explanations for these differences are partially attributed to differences in individuals' travel attitudes, capabilities, and preferences (De Vos et al., 2025; van der Vlugt et al., 2022). In addition, objective measures often rely on travel time as the only impedance factor, while other factors that might also shape individuals' perceptions of the ease or difficulty of reaching destinations are not considered, such as safety, comfort, and even enjoyment of the trip (Handy and Clifton, 2001; Handy, 2020).

Another factor that might impact the relationship between objective and perceived accessibility is adaptive preference, in which political oppression, poverty, and adverse social conditions distort the preferences of individuals living in deprived circumstances (Baber, 2007). In this situation, someone with restricted financial resources living in an area with low (objective) accessibility might adapt their expectations and preferences to align with their circumstances, and might therefore not consider their accessibility levels to be insufficient (Negm et al., 2025).

Residential self-selection can also impact the relationship between perceived and objective accessibility, in which people may choose their residential locations based on travel needs, abilities, and travel-related attitudes (Cao et al., 2009). In fact, residential location decisions differ across income groups, impacting the relationship between objective job accessibility and employment status (Hu, 2017). For instance, high-income individuals often prioritize living in places with low job accessibility because they prefer suburban housing amenities, such as green spaces and larger homes, over proximity to work, accepting longer commutes because they can afford - and might even prefer to have - a car. As a consequence, their perceived ability to reach employment opportunities may be defined less by their neighbourhood's job accessibility and more by their resources and preferences. On the other hand, low-income individuals might prefer to live close to employment hubs to maximize their chances of having a job, while reducing commuting and housing costs.

Objective accessibility is still commonly used to evaluate the effect that the interaction between land-use characteristics and the transportation system has on perceived job accessibility and on employment outcomes. In addition, existing studies on perceived accessibility seem to have difficulties in creating standardized definitions and measurement tools, contributing to the somewhat limited knowledge on the topic (Negm et al., 2025). In this context, connecting objective to perceived accessibility helps to capture non-spatial and employment-related barriers that are not visible in place-based measures alone, since the objective

accessibility might be perceived differently even between people living within that same neighbourhood. In this paper, we define sufficient levels of accessibility based on the relationship between objective accessibility and people's perception of job accessibility and employment outcomes, while also controlling for individual characteristics.

2.2. Job accessibility and employment outcomes

Studies connecting employment outcomes to job accessibility date back to the 1960s, when John Kain (1968) proposed the spatial mismatch hypothesis while analyzing Detroit and Chicago, USA. The author hypothesized that the low employment rates among Black Americans in inner cities resulted from employment decentralization and the difficulty faced by the Black population in moving out of these areas, primarily due to racial discrimination and segregation.

Since Kain (1968), studies linking objective job accessibility and different employment outcomes have presented mixed results. Bania et al. (2008) found no significant effect for job access when examining its relationship with employment, earnings, hourly wage, and weekly hours worked for women leaving welfare in Ohio, USA. Delmelle et al. (2021) also found that changes in car accessibility had no significant impact on employment rates for different population income groups in the Metropolitan Area of Charlotte, North Carolina, USA, arguing that the spatial separation between lower-wage workers and their workplaces might already be relatively small. Studying Los Angeles, USA, Merlin and Hu (2017) found mixed-effects of job accessibility on employment status, with a stronger association for more highly educated groups. The authors argued that lower-educated groups may be more limited by non-spatial factors that impede their employment opportunities, such as qualifications, labour market readiness, and a lack of social networks. On the other hand, for workers with a higher level of education, the time and monetary costs of spatial separation itself may play a relatively more important role, so accessibility measures may have a greater impact on their decisions and ability to enter the labour market.

In contrast, other studies found significant effects of job accessibility on employment outcomes (Bastiaanssen et al., 2021; Duarte et al., 2023; Jin and Paulsen, 2018; Johnson et al., 2017; Matas et al., 2010). Jin and Paulsen (2018) used a difference-in-differences model for the Chicago Metropolitan Area, finding that increasing job accessibility between 2010 and 2017 was associated with reduced unemployment rates. Duarte et al. (2023) found that job accessibility had a positive effect on participation in the labour market for women, a negative effect on informal employment for women, and a positive effect on the probability of being employed for both women and men. More recently, Bastiaanssen et al. (2025) examined differential employment effects of job accessibility by public transport in combination with the bicycle for different educational groups at the national level in the Netherlands. They found that jobs for lower-educated groups are often located outside areas of high accessibility, thereby reducing job accessibility among these groups, while jobs for higher-educated groups tend to be concentrated in and around downtowns and are associated with greater accessibility. In this study, the assumed connection between the spatial dimension of accessibility and people's perceived accessibility and employment outcomes is the basis for the identification of equity standards.

2.3. Equity standards

An equity standard, when operationalized, defines the level of fairness to be adopted and the acceptable inequality (Soukhov et al., 2025b). Accessibility standards typically combine a concept of fairness with threshold criteria, reflecting both vertical and horizontal equity approaches. Vertical equity is often operationalized through opportunity-based standards defined for specific population groups (e.g., minimum access to opportunities within a 10-minute walking trip for older adults), whereas horizontal equity commonly appears in the use of uniform travel impedance thresholds applied across all potential users (e.g., a maximum acceptable distance to a transit stop).

Yenisetty and Bahadure (2020) set an opportunity-based standard, considering a distance of more than 1,200 meters between amenities, services, and facilities by public transport to be unfair when analyzing cities in India. Tiznado-Aitken et al. (2018) defined unfair cases as those where bus stops and metro stations are located more than 1 kilometer away. Sun and Thakuriah (2021) adopted 400 meters as the maximum

acceptable walking distance to bus and tram stops, and 800 meters to ferry stations in England. In another study, Peungnumsai et al. (2020) identified opportunity standards through the ratio of public transport supply to demand, considering conditions fair when demand is greater than supply.

The use of different thresholds reflects the idiosyncratic manner in which standards are developed, and the rationale for choosing a certain distance to, for example, bus stops or train stations, is often unclear. In addition, the use of distance-based thresholds may be problematic, since the time required to cross a certain distance can vary considerably within a city, even for the same mode of transportation. To address this problem, we suggest standards based on key thresholds of objective accessibility.

However, beyond making explicit how standards are defined, it is also important to make transparent the moral principles that guide their definition, because different theories of justice lead to different perspectives on how policies should manage the distribution of transport-related benefits. Pereira and Karner (2021) argue that transportation equity research benefits from linking two justice theories: Rawlsian egalitarianism (Rawls, 1999) and the Capability Approach (Sen, 1979, 1999, 2005, 2009). Rawls' Difference Principle holds that inequalities are only justifiable if they benefit the least advantaged members of society, centred on the fair distribution of primary social goods - resources and opportunities that every person needs to pursue their life plans. Sen builds on this, arguing that the focus should shift from primary goods to human capabilities: the opportunities available to individuals and their freedom to choose and act, recognizing that the same resource can translate into different real opportunities depending on individual circumstances. Within transportation research, accessibility to opportunities can therefore be understood as a capability that enables participation in social, economic, and community life (Pereira et al., 2017; Luz and Portugal, 2022).

Both theories inform two moral perspectives commonly applied, though rarely made explicit, in transportation equity research (Martens, 2016a; Wee and Geurs, 2011). The sufficientarian perspective asks why some groups or areas lack sufficient access, building on the idea of a minimum threshold of accessibility required to meet basic needs, with priority given to those falling below it. The egalitarian perspective instead focuses on relative differences in accessibility between groups or areas, treating unequal access as unfair regardless of whether a group's absolute access level meets a minimum threshold (Pereira and Karner, 2021). Applying either perspective therefore requires identifying which groups are of concern: the sufficientarian view asks who falls below the threshold, while the egalitarian view asks between whom access differs.

In practice, this attention has centred on disadvantaged population groups, including those affected by low income, limited ability, and historical discrimination, which have become the focus of transportation research (Lucas, 2019). This is due to the potential of historical discriminatory policies and other disadvantages to negatively impact their transport experiences (Pereira and Karner, 2021). Additionally, these groups have often been excluded from the transportation planning process, resulting in their needs frequently being not recognized. Within studies that investigated associations between objective accessibility and employment outcomes, relevant groups include youth, adults, women, children, low-income individuals, zero-vehicle households, people with disabilities, single-parent households, refugees, and racialized populations (Bania et al., 2008; Hu, 2017; Middleton and Spinney, 2019; Tabascio et al., 2024).

In this study, we recognize accessibility as a human capability (Sen, 2005, 1999; Pereira et al., 2017; Luz and Portugal, 2022), something required for a dignified life (Karner et al., 2020; Dos Santos et al.), and one of the main benefits of transportation to be distributed across society in order to increase people's freedom. Following the sufficientarian perspective, we define equity standards as minimum accessibility thresholds required to reduce the likelihood of negative perceptions and outcomes. Following an egalitarian view, these thresholds are defined with particular attention to transport-disadvantaged population groups. In doing so, the paper presents an analysis that identifies minimum acceptable levels of accessibility, discusses inequities in these values, and presents a generalizable methodology for other geographical areas, populations and opportunity types.

3. Materials and methods

Our approach for obtaining equity standards of accessibility based on the relationship between objective measures, people's perceptions, and employment outcome can be synthesized in the following main steps:

modelling job accessibility, predicting individual perception and outcome using objective accessibility, setting equity standards for transportation, measuring transportation equity by using Foster-Greer-Thorbecke (FGT) measures (Foster et al., 2010).

3.1. Study area and data sources

The study area for this research is the metropolitan region of the Greater Toronto Area and the Hamilton Area (GTHA), in Ontario, Canada. More specifically, the study area refers to the Census Divisions of the regional municipalities of Hamilton, Halton, Peel, Toronto, York, and the Durham subdivisions within the Toronto Census Metropolitan Area (Figure 1). According to the latest census (Canada, 2023), over 7.2 million residents live in this area, which is spread across around 7,300 square kilometres.

Regarding the primary data sources, this study uses data from two different sources: the latest Canadian Census (2021) from Statistics Canada to calculate objective accessibility; and the National Survey of Transport Poverty (NSTP) from the Mobilizing Justice Partnership (Tiznado-Aitken et al., 2025) to obtain variables related to individuals' perceived accessibility and employment outcomes. The Census was selected due to its various socio-economic and demographic variables and due to the "Commuting to Work" questionnaire, allowing the identification of individuals in the labour force and including variables required to calculate job accessibility, such as place of work, commuting modes, and commuting duration. The Census also provides standardized spatial datasets with statistical and administrative geographical boundaries, including the Census Tract (CTs), which are used in this study as the spatial units used for calculating accessibility measures (obtained using the {cancensus} `R` package (von Bergmann et al., 2021)). According to the Census (Canada, 2023), 3,868,120 individuals in the labour force reside in the study area, competing for a total of 3,043,730 job opportunities.

The Census also provides the number of people in low income, which we use to assess accessibility gaps among low-income residents. Low income is defined using the Low-Income Measure, After-Tax (LIM-AT), a relative threshold set at 50% of the median adjusted after-tax household income, where each household's after-tax income is adjusted by dividing it by the square root of household size. According to the 2021 Census, around 10% of the study area population lives in low income.

The second source of data, NSTP (Tiznado-Aitken et al., 2025), is a large-scale survey designed to understand TRSE and transport poverty in Canada. It is a representative sample of more than 27,000 people aged 15 years and older across the country, and it provides an unprecedented look at the transportation challenges that Canadians face, especially among transport-disadvantaged groups whose experiences are often overlooked. The survey is openly available to researchers, decision-makers, and the public, and offers information on household dynamics, respondents' perceptions, and individual-level socioeconomic data. The NSTP provides respondent weights to enable the creation of statistics that are consistent and representative of the Canadian population. From this survey we select variables corresponding to respondents' perceptions of accessibility and employment outcomes, in addition to socioeconomic and demographic variables at the individual level. The responses are geocoded by place of residence and can be assigned to Census geographies to link them to objective accessibility.

For this study, only a sample of the NSTP was used. This sample corresponds to labour force respondents, whether employed or unemployed, residing in the GTHA who were asked, and answered, at least one of the two questions used as dependent variable. Using a continuum scale from 0 (completely disagree) to 100 (completely agree), the respondents answered to what extent they agreed with the statements in Table 1, namely *"There are too few jobs available for me in my ideal commuting time"* and *"I have had to decline employment opportunities because of my transport situation"*. In total, the NSTP sample refers to 2,637 respondents that represent 4,233,447 individuals in the labour force (weighted sample).

Since respondents could skip either of the questions, the two dependent variables are not observed for the same individuals. Instead of restricting the analysis to respondents who answered both questions, each model uses all respondents who answered the corresponding question and have complete information on the explanatory variables: 2,480 respondents (representing 4,013,199 individuals) in the perceived accessibility model and 2,365 respondents (representing 3,859,826 individuals) in the declined job offers model. The descriptive statistics in Tables 1 and 3 refer to the combined sample of respondents who answered at least one of the two questions.

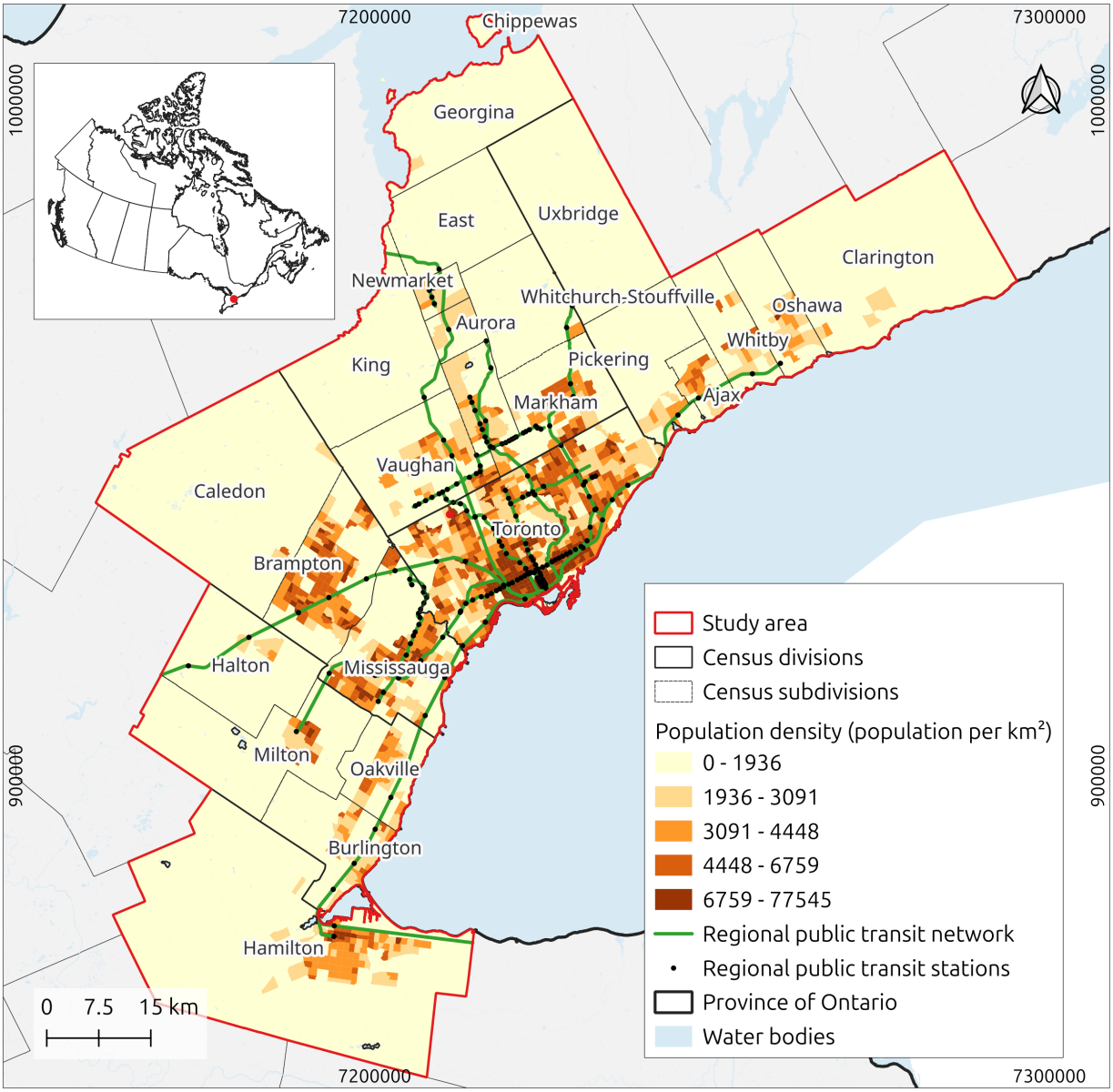


Figure 1: Map of the study area (population density is classified using quintile breaks).

Table 1: Weighted summary statistics of the dependent variables.

To what extent do you agree with the following statements?	**N = 4,233,447**
There are too few jobs available for me in my ideal commuting time	58.64 (27.69)
I have had to decline employment opportunities because of my transport situation	40.10 (31.22)

[1] Mean (SD)

3.1.1. Objective accessibility inputs: labour force, job opportunities, and travel costs

Calculating objective accessibility, as operationalized in this paper (see following section), requires three inputs: the number of opportunities at each destination, the population competing for them at each origin, and the cost of travelling between origins and destinations, which is translated into travel impedance through a mode-specific function.

Using microdata from the Census, we calculated for each CT the number of people in the labour force that commute to work by walking, cycling, private vehicles (those who commute by car, truck, or van as a driver or passenger, and motorcycles or scooters), and public transit (those who commute by bus, subway, light rail, streetcar, commuter train, and passenger ferry). The labour force is defined as in the Canadian Census: civilians aged 15 and older who are employed or unemployed. We distributed remote workers according to the share of each transportation mode in each CT. Then, we summed the number of jobs in each CT considering the place of work mentioned by each worker.

We calculated the travel times between origins and destinations for each travel mode using the {r5r} `R` package (Pereira et al., 2021) with a street network obtained from OpenStreetMap. We used the General Transit Feed Specification (GTFS) obtained from the Canadian Public Transit Network Database (Canada, 2025) to retrieve the public transit schedule and coverage. Considering that our study area contains different transit agencies, we evaluated the GTFS files to identify a common date/week with public transit service. After this analysis, we set 8 a.m. (morning peak time) of December 22, 2025, as the trip departure time. Although this date falls during Christmas week, there were no differences in the public transit service schedules of the transit agencies due to the holiday, which makes this date representative of a regular weekday in terms of public transit travel time estimation. We also used Digital Elevation Models (DEM) from the Canadian Digital Surface Model (CDSM), with a spatial resolution of about 20 meters, to estimate the effect of declivity on walking and cycling times. For cycling trips, we defined a standard level of traffic stress (LTS) as tolerable for most mainstream adults. LTS defines the level of comfort or danger that a street presents for cyclists based on speed limits, lane counts, and infrastructure. The standard LTS defined in {r5r} (Pereira et al., 2021) includes bike lanes with physical separation from multi-lane or higher-speed traffic. We set a maximum duration of 180 minutes (3 hours) for private vehicles and public transit trips, with a maximum walking time of 5 minutes to reach the vehicle and 15 minutes to reach the public transit stop/station; for walking and cycling trips, we set a maximum duration of 90 minutes (one and a half hours). After estimating the travel times between CTs, we assumed that all intra-CT trip durations are equal to 0.1 minute.

Commute duration was used to calibrate Probability Density Functions (PDFs) to represent travel impedance, following the approach of Soukhov and Páez (2024), Verma and Ukkusuri (2025), and Dos Santos et al. (2026), where the impedance value represents the probability that the trip will occur given a specific travel time. We used the {fitdistrplus} `R` package (Delignette-Muller and Dutang, 2015) and tested different functional forms (uniform, negative exponential, gamma, normal, and lognormal distributions) to select the best PDF for each mode of transportation and municipality, using as the selection criteria the lowest Akaike Information Criterion (AIC) value (Akaike, 1974). The estimated impedance functions appear in Appendix A.1.

3.2. Calculating objective accessibility

In this study, we implement the multimodal spatial availability measure of Soukhov et al. (2024). This measure is a constrained indicator of accessibility that can account for multiple modes (see Soukhov et al., 2023, 2025a). This measure considers competition for opportunities, which prevents a single job opportunity from being available to more than one person. Competitive measures matter when considering employment outcomes because jobs are exclusive, and their use by one person prevents their use by another. In addition, this measure relies on a proportional allocation that balances the cost of travel and the population size, imposing a constrained calculation that allocates the opportunities to each origin. This ensures that, when all accessible jobs are summed, the sum is equal to the total number of opportunities in the region. Lastly, this measure is a more interpretable indicator since it returns scores in the units of opportunities.

We calculated the multimodal spatial availability V_i^m using Equation (1):

$$V_i^m = \sum_{j=1}^{J} O_j \; F_{ij}^{tm} \tag{1}$$

where:

- V_i^m is the spatial availability of opportunities at origin i and travel mode m;
- O_j is the number of opportunities at destination j;
- F_{ij}^{tm} is a balancing factor allocating opportunities at j to origin i considering mode m.

F_{ij}^{tm} incorporates both population demand and travel impedance, defined according to Equation 2:

$$F_{ij}^{tm} = \frac{F_i^{pm} \cdot F_{ij}^{cm}}{\sum_{m=1}^{M} \sum_{i=1}^{N} F_i^{pm} \cdot F_{ij}^{cm}} \tag{2}$$

The population-based allocation factor is obtained by:

- $F_i^{pm} = \frac{P_i^m}{\sum_m \sum_i P_i^m}$

Where P_i^m represents the labour force at origin i using mode m competing for the opportunities O at the destination j.

In turn, the travel impedance allocation factor is given by:

- $F_{ij}^{cm} = \frac{f^m(c_{ij}^m)}{\sum_m \sum_i f^m(c_{ij}^m)}$

Where:

- c_{ij}^m is the cost to travel between i and j for mode m;
- $f^m(c_{ij}^m)$ is an impedance function representing the effect of travel cost.

3.3. Modeling the relationship between objective accessibility, perceived accessibility, and employment outcomes

In this study, we aim to define minimum levels of objective accessibility by identifying key thresholds in its relationship with perceived accessibility and employment outcomes. The underlying hypothesis of this strategy is that changes in the land-use and transport characteristics of a neighbourhood, captured and represented by objective accessibility, lead to changes in how residents perceive their ability to reach job opportunities and in their reported employment outcomes. We hypothesize that residents of neighbourhoods with more (objective) accessibility are less likely to perceive job scarcity or to report having declined job offers due to transport constraints, independent of their individual demographic and socioeconomic profiles. In our approach, we use objective accessibility not as a causal exposure in itself, but as a proxy for the underlying land-use and transport conditions that, as we hypothesize, shape individual perceptions and outcomes.

Following the steps suggested by Huntington-Klein (2021), we draw a Directed Acyclic Graph (DAG) to graphically represent our hypothesis (Figure 2). DAGs show a chain of hypothesized causal effects by using nodes to symbolize random variables and arrows to represent the causal effect between two random variables in the direction of the arrow (Cunningham, 2021; Huntington-Klein, 2021).

We hypothesize that there is an association between objective accessibility, perceived accessibility, and employment outcomes. However, there are other ways in which they are connected, represented by the other nodes of the diagram. First, there are mediator components that influence the way a person perceives, or make use of, objective accessibility. Each individual perceives the same neighbourhood characteristics differently based on their capabilities and preferences, as well as their travel options and experiences. For instance, a person with a disability that prevents or makes it difficult for them to walk or cycle will probably

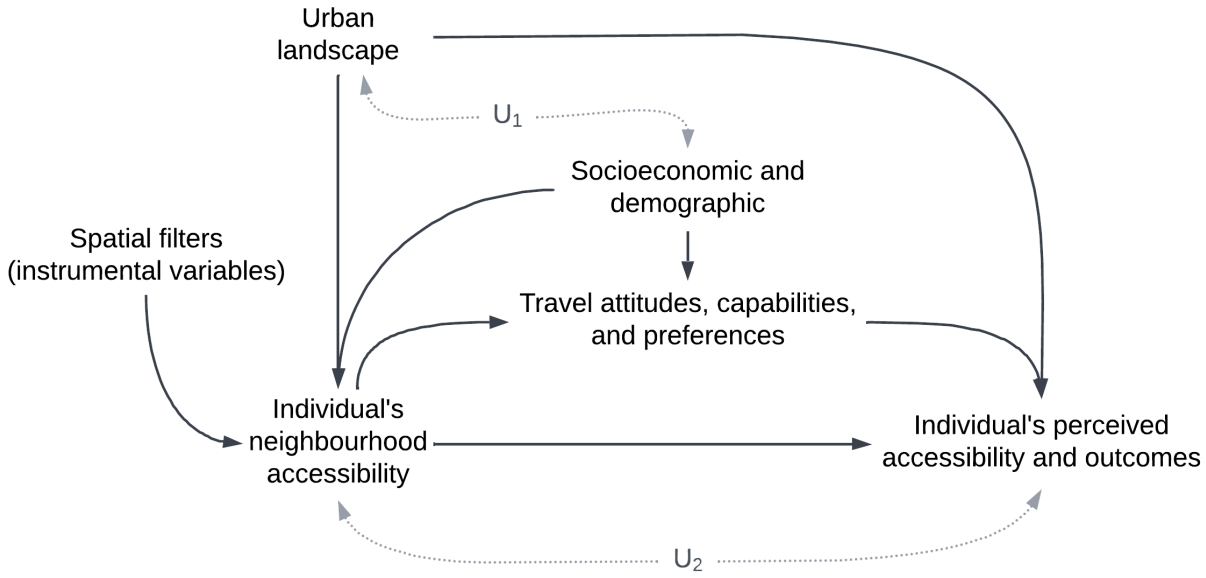


Figure 2: Directed acyclic graph (DAG) of objective accessibility on perceived accessibility.

not enjoy the benefits of active accessibility to the same extent as a person without such a disability (Church and Marston, 2003); someone without a car will not take advantage of high car accessibility; someone without a monthly transit pass might take less public transit than someone with a transit pass, therefore reducing their potential for spatial interaction promoted by higher public transit accessibility.

In addition, place of residence, and consequently an individual's neighbourhood accessibility, is also influenced by socioeconomic factors, with household income as the main driver of this decision, and also by demographic characteristics (such as gender, race, citizenship, among others). These socioeconomic and demographic factors not only influence where someone will live, but also shape how individuals experience travel and, consequently, make use of their neighbourhood accessibility.

However, socioeconomic and demographic factors and the urban landscape of the place of residence are also linked through a common cause (U_1) that we cannot measure directly, such as generational wealth, which likely influences both someone's neighbourhood and their income level. Other unmeasured factors, such as residential self-selection and adaptive preference, likely affect objective accessibility, perceived accessibility, and employment outcomes (U_2).

Because other paths link objective accessibility to perceived accessibility and employment outcomes, the observed association between them cannot be assumed to reflect the true effect of accessibility on how people experience and evaluate their access to employment opportunities. If objective accessibility is to be used as a benchmark or standard for equitable transportation planning, we believe it is important to apply methods that aim to isolate its association with perceived accessibility and employment outcomes, after accounting for these confounding mechanisms, instead of relying on a raw statistical association that may be inflated, offset, or masked by unobserved individual preferences and constraints. Our approach relies on, first, controlling for individual-level covariates capturing demographic, socioeconomic, and travel-related characteristics; and second, applying a framework using instrumental variables.

Guided by the diagram and the supporting literature, we identified control variables corresponding to each node, following standard causal-diagram identification methods (Cunningham, 2021; Huntington-Klein, 2021). We selected the following variables:

- Objective accessibility (treatment): multimodal spatial availability per person by transportation mode.

Table 2: Quintile thresholds used to categorize dependent variables.

Outcome	20%	40%	60%	80%	100%
Few jobs available in my ideal commuting time	35	52	70	86	100
Declined job offers because of my transport situation	8	24	51	71	100

- Perceived accessibility (outcome 1): agreement with *"there are too few jobs available for me in my ideal commuting time,"* obtained from NSTP.
- Self-reported employment outcome (outcome 2): agreement with *"I have had to decline employment opportunities because of my transport situation,"* obtained from NSTP.
- Travel attitudes, capabilities, and preferences: most used transportation mode; whether the preferred transportation mode matches the most frequent mode; whether the individual has a monthly or annual public transit pass; number of cars and bikes in the household; and whether the person has any disability, all obtained from NSTP.
- Socioeconomic and demographic: whether the person is Indigenous or from a racialized population group; whether the person self-identifies as a woman; or as a transgender or non-binary person; age; household size and household income; and citizenship (temporary visa or refugee), all obtained from NSTP.
- Urban landscape: population centre class (rural, small, medium, or large urban population centre) and population density (dissemination area level), obtained from Census.

In total, we estimate two regression models, one for each dependent variable: perceived accessibility and declined job offers. Both models use the instrumental variable approach described in the following subsection, and each includes the three accessibility indicators (active, public transit, and private vehicle spatial availability per person), the interactions between active and transit accessibility and their respective user indicators, and the full set of control variables at the individual and household levels from the NSTP.

Although respondents expressed their level of agreement on a continuous scale from 0 to 100, we treat these responses as ordinal measures, not interval measures. Agreement scales based on sliders do not guarantee that equal numerical differences reflect equal differences in perception: a change from 10 to 20 may not represent the same variation in agreement as a change from 80 to 90. Furthermore, respondents may use the scale in different ways, for example, by grouping their responses around the extremes or the midpoint. According to Liddell and Kruschke (2018), analyzing ordinal responses as if they were metric can distort the estimated effects and inflate error rates. We can, however, have more confidence in the order of the responses, since a respondent who selects 80 agrees more strongly than one who selects 60.

Therefore, using respondent weights, we transformed the dependent variables into an ordered Likert scale based on their quintile position, categorizing them as: "Most Negative", "Negative", "Neutral", "Positive", and "Most Positive". The thresholds used for this categorization are presented in Table 2. Since greater agreement with either statement indicates a worse situation, the highest quintile of agreement corresponds to "Most Negative". Quintiles also ensure a sufficient and balanced number of respondents in each category, which is needed for stable estimation of the ordinal models, and provide a middle category that serves as the neutral reference for the dependent variable.

Considering that the outcome is an ordinal categorical variable, we used weighted ordinal logistic regressions (Lumley, 2011) to model the relationship between the objective accessibility indicators and the categories of the dependent variables, using Equation 3:

$$\log\left(\frac{P(Y \leq j)}{P(Y > j)}\right) = \theta_j - (\beta_1 X_1 + \beta_2 X_2 + ... + \beta_p X_p) \tag{3}$$

where Y represents the response category of the respondent (ranging from "Most Negative" to "Most Positive"), and $X_1, X2, ..., X_p$ represent the set of explanatory variables, and θ_j and β_p are unknown parameters to be estimated. The parameters β_p are constant for all outcome categories and measure the change in

Table 3: Weighted summary statistics of the explanatory variables, based on the NSTP sample respondents who answered at least one of the two dependent-variable questions.

Explanatory variables	**N = 4,233,447**
Active job accessibility per active traveller	0.06 (0.12)
Transit job accessibility per transit user	0.14 (0.09)
Private vehicle accessibility per user	1.00 (0.27)
Active commuter	676,577 (16%)
Public transit user	926,245 (22%)
Person with disability	825,488 (19%)
Number of Cars	1.35 (0.91)
Number of Bikes	1.21 (1.35)
Preferred mode match most frequent mode	3,221,068 (76%)
Indigenous or racialized person	2,200,448 (52%)
Temporary visa or refugee	310,758 (7.3%)
Woman, transgender or non-binary person	2,040,872 (48%)
Age	40.35 (13.55)
Household size	2.93 (1.57)
Household income	
Less than $40,000	573,963 (14%)
$40,000 to $80,000	1,161,619 (27%)
$80,000 to $120,000	944,571 (22%)
More than $120,000	1,553,296 (37%)
Population density (people/km2)	7,592.12 (12,899.53)
Population centre class	
Rural	98,477 (2.3%)
Small	59,944 (1.4%)
Medium	64,705 (1.5%)
Large	4,010,321 (95%)
Public transit availability in the neighbourhood	4,056,243 (96%)
Monthly or annual public transit pass	1,196,048 (28%)

[1] Mean (SD); n (%)

the log-odds of reporting higher response categories associated with each explanatory variable, while θ_j represents the intercept separating the response categories.

Because of the high correlation between public transit and bicycle accessibility (with Pearson correlation above 0.80), and because only 61 NSTP respondents (less than 3% of the sample) mentioned cycling as their main commuting mode, we combined walking and cycling into a single active transportation indicator, summing the jobs spatially available by each mode. Including bicycle accessibility as a separate indicator, given this small subsample and high collinearity with transit accessibility, would likely have caused convergence problems when creating our ordinal models. Fitting these models also requires a sufficient number of respondents within each category and combination of categories, which the small cycling subsample could not support. However, we acknowledge that this combined active measure does not account for the difference in travel speed between walking and cycling, which can be substantial, and may understate the accessibility gains cycling offers relative to walking over longer distances.

Table 3 presents the variables included in the regression models and their descriptive statistics for the NSTP sample (weighted mean and weighted standard deviation for continuous and dummy variables, and weighted count and percentage for categorical variables, created using the {gtsummary} `R` package (Sjoberg

et al., 2021)).

3.3.1. Instrumental variables

The presence of U_2 in our DAG implies the possibility of bias caused by omitted variables that simultaneously affect objective accessibility, perceived accessibility, and employment outcomes - a source of endogeneity that compromises the consistency of the estimators (Antonakis et al., 2010; Cunningham, 2021; Huntington-Klein, 2021). We address this using an instrumental variable (IV) technique, where an instrument Z must be correlated with the outcome only through objective accessibility. Examples of instrumental variables used in studies analyzing the relationship between employment outcomes and job accessibility include Euclidean distance from the spatial unit centroid to the nearest major road or employment subcenter (Delmelle et al., 2021; Jin and Paulsen, 2018), share of automobile ownership (Hu, 2017), transit accessibility and car unavailability (Johnson et al., 2017), distance to rivers (Duarte et al., 2023; Luz and Portugal, 2022), and population density (Bastiaanssen et al., 2021, 2025).

Two challenges arise in the search for relevant instruments: the likelihood of correlation between instruments and the error term increases with the strength of their correlation with the endogenous variable, while instruments only weakly correlated with the endogenous variable tend to perform poorly. Le Gallo and Páez (2013) developed an approach that relies on the use of synthetic instruments derived from the spatial configuration of the data (see Getis and Griffith, 2002; Griffith, 2003), which ensures high correlation with the treatment while, potentially, remaining exogenous to the outcome. Following this approach:

1. We built a queen-contiguity matrix for the census tracts to represent spatial weights, expressing the existence of a neighbouring relation as a binary relationship (1 for neighbours and 0 otherwise).
2. Then, we computed all eigenvectors associated with the contiguity matrix. These eigenvectors are orthogonal to one another and represent a portion of the variance in the spatial contiguity matrix.
3. After this, we initialized a variable S with value 0 and linearly regressed each accessibility indicator (the independent variable) on each eigenvector and S. If the eigenvector coefficient was significant (adopting a p-value ≤ 0.05), we multiplied the eigenvector by its coefficient and added it to the value of S. After this process, S represents the spatial filter associated with the accessibility indicator.

All spatial filters presented a Pearson correlation above 0.88 with their respective accessibility measures and absolute correlations lower than 0.15 with the original continuous dependent variables (before transforming them into categorical variables), consistent with the relevance and exclusion conditions required for valid instruments. In addition, we performed Wald tests to evaluate the null hypothesis of weak instruments and insufficient explanatory power to predict objective accessibility, and Wu-Hausman tests to assess whether the accessibility indicators are endogenous. For the ordinal models, endogeneity is assessed through the statistical significance of the first-stage residuals in the second-stage regressions, which in the two-stage residual inclusion framework described below is a Hausman-type test of exogeneity (Terza et al., 2008; Wooldridge, 2010). As a robustness check, we also report weak-instrument and Wu-Hausman tests for linear models estimated on the original continuous dependent variables (Appendix A.2).

Since our dependent variables are not linear, the popular technique of replacing endogenous variables by their first-stage predictors (two-stage least squares estimation) is not appropriate - the so-called "forbidden regression" in econometrics (Wooldridge, 2010; Cunningham, 2021; Huntington-Klein, 2021). Therefore, we applied the two-stage residual inclusion (2SRI) approach (Terza et al., 2008), in which, in the first stage, we estimated the accessibility indicators using a linear regression including all control variables plus spatial filters (our instrumental variables). Next, in the second-stage, the dependent variables were estimated using the accessibility measures, the control variables, and the residuals from the first-stage regressions. Both stages used respondent weights. In this approach, we use the regular treatment to predict the outcome, but also control for the unexplained part of the treatment.

3.4. Identifying equity standards

Considering only the cases in which the objective accessibility was statistically significant and associated with a higher Likelihood of a positive response, we measured changes in the predicted probabilities across

response categories by varying the levels of objective accessibility, while controlling for the other explanatory variables. There are many possible ways of controlling the explanatory variables (e.g., fixing the number of cars in the household as 0, 1, or 2+; or fixing the respondent age as 30, or 31, and so on). Based on this, we decided to evaluate the effects of objective accessibility in three distinct individual profiles:

- A transport-disadvantaged profile, with individual-level control variables set to the values corresponding to populations facing the greatest transportation challenges.
- The most prevalent profile, reflecting modal characteristics of the study area population and with control variables set to their most common values; and,
- An advantaged profile, defined by selecting for the most privileged population segment in terms of their transport experience.

For each individual profile, we increased the value of each accessibility measure by 0.01 jobs per person, while holding constant other accessibility measures (representing alternative transportation modes) at their weighted median values. We also adopted the weighted median value for the population density and a constant zero value for the residuals obtained from the first-stage regressions. For each transportation mode, we evaluated the effect of accessibility by considering the individual profiles as users of that specific mode. That is, for active travel accessibility, all individual profiles were set as active travellers, and so on for the remaining modes.

Then, for all individual profiles, we identified the minimum value of objective job accessibility at which the combined probability of the non-negative response categories ("Neutral", "Positive", and "Most positive") exceeds the combined probability of the negative categories ("Negative" and "Most negative"). Because the five probabilities sum to 100%, this is equivalent to the probability of a negative response falling below 50%. We name this value the conservative threshold. Next, we defined the progressive threshold as the minimum accessibility value at which the combined probability of the positive categories ("Positive" and "Most positive") exceeds the combined probability of the negative categories, excluding the neutral category from both sides. Finally, we set the intermediate threshold as the midpoint between the conservative and progressive thresholds.

We base our methodology on egalitarian and sufficientarian perspectives, and we propose defining equity standards based on the transport-disadvantaged profile. Our approach to operationalizing these justice principles relies on identifying the minimum accessibility level (sufficientarianism) at which the most disadvantaged profile (egalitarianism) is more likely to give a non-negative response than a negative one (conservative standard), and more likely to give a positive response than a negative one (progressive standard).

3.5. Measuring transportation equity

We calculated the extent and severity of accessibility gaps using FGT measures (Foster et al., 2010), according to the Equation 4. FGT measures evaluate the number of people below a standard and the size of the accessibility gap relative to it.

$$FGT_{\alpha} = \frac{1}{N} \sum_{H}^{i=1} \left(\frac{z - y_i}{z} \right)^{\alpha} \tag{4}$$

Where N is the size of the total population; z is the accessibility standard; H is the number of individuals below the z; y_i is the accessibility level of each individual i; and α: determines how sensitive the index is to changes in the accessibility gap.

Changing the values of α enables the calculation of different FGT indices (Karner et al., 2024). When $\alpha = 0$, FGT_0 measures the extent of undersupply as a simple headcount, indicating the proportion of people below the standard. When $\alpha = 1$, FGT_1 measures the severity of the undersupply as the average percent distance between the standard and the accessibility levels of the individuals below it.

We applied the FGT measures to the general population in each CT, and also to the population classified as low-income according to the Census, as defined by the LIM-AT.

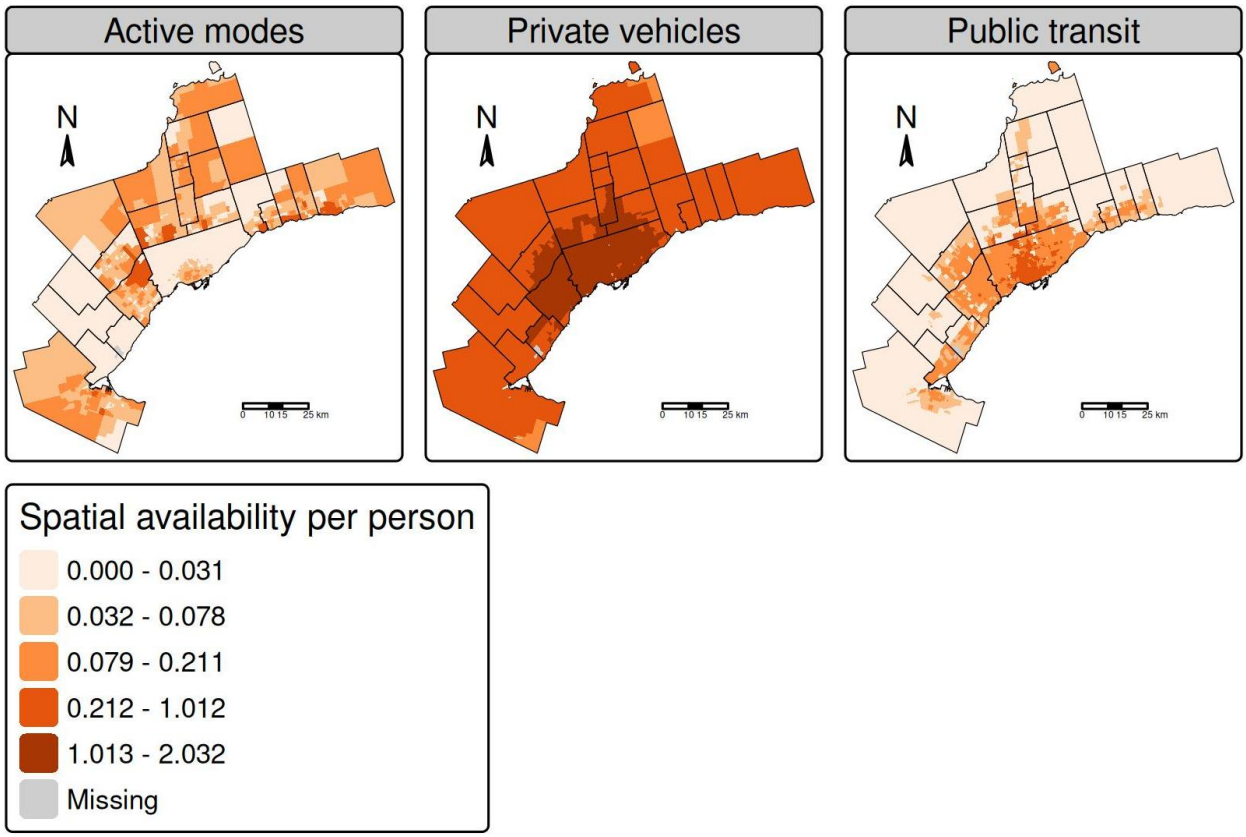


Figure 3: Maps of the multimodal spatial availability per person, considering active, public transit and private vehicles (accessibility values are classified using quintile breaks).

4. Results and discussions

4.1. Multimodal spatial availability per person.

Figure 3 presents the maps with the multimodal spatial availability of jobs per person. These maps show how, generally, private vehicle users can access more job opportunities, but also show that there is not much spatial variation when compared to the other two modes, with most of the CTs in the study area classified in the second-highest or highest accessibility categories. Toronto and Mississauga present the highest level of private vehicle accessibility.

Transit accessibility appears to be more spatially distributed across the study area, with the city of Toronto presenting the highest rate of jobs accessible per transit commuters. Overall, we can observe an expected spatial pattern where CTs close to public transit infrastructure present higher levels of transit accessibility when compared to suburbs and rural areas, which often lack public transit services.

Next, the map of the active jobs accessible per active traveller (walkers or cyclists) does not present a clear spatial pattern, with many suburban and rural areas presenting higher levels when compared to more urbanized areas. This can be explained by either a higher number of jobs accessible within typical walking or cycling trip durations, less probable; or, probably, because the number of active travellers in these areas is very low, resulting in a higher rate of jobs per person, in a way that higher accessibility values in these areas may reflect low competition rather than higher absolute job availability.

Although private vehicle commuters represent about 78% of the labour force, they compete for 96% of the jobs in the study area. Transit users represent 15% of the labour force, but they reach only 3.4% of

the jobs available, while the remaining 0.6% of jobs available are competed for by the 7% of active users. On average, each active user respondent from the NSTP can access 0.06 jobs, while each transit user can access, on average, 0.14 jobs. Both these measures are substantially lower than the number of jobs that each private vehicle commuter can reach, which is approximately one job per commuter on average.

4.2. The impact of objective accessibility

Table 4 presents the two probability models linking objective accessibility to perceived accessibility and employment outcome. Based on Wald tests, we reject the null hypothesis of weak instruments and insufficient explanatory power to predict our endogenous variables (the objective accessibility indicators), as all F-statistics are well above the conventional rule-of-thumb value of 10 and statistically significant (p-value < 0.05, Appendix A.2).

The evidence of endogeneity differs between the two models. In the perceived accessibility model, the residual from the first-stage of accessibility by private vehicle is statistically significant (OR = 1.17, p = 0.014), indicating that this indicator is endogenous to perceived accessibility. The Wu-Hausman test applied to the linear version of this model is also significant (F = 2.32, p = 0.041; Appendix A.2). The positive coefficient of this residual means that unobserved factors that make respondents to live in neighbourhoods with greater accessibility by private vehicle than predicted by the instruments and controls are also associated with more positive perceptions, consistent with the theory of residential self-selection, in which respondents with unobserved preferences for car travel may choose neighbourhoods that are accessible by car and, for the same reasons, perceive more jobs within their ideal commute time. The residuals for accessibility by public transportation and active transportation are not statistically significant; therefore, we find no evidence that these indicators are endogenous. In the declined job offers model, neither the residuals from the first stage nor the Wu-Hausman test (F = 1.14, p = 0.338; A.2) indicate endogeneity. Even so, we decided to keep the instrumental variables framework in both models to allow for a direct comparison between them.

Objective accessibility indicators were partially significant across the models. Private vehicle accessibility did not present significant values for any of the models, probably due to its ubiquity and homogeneity across the study area.

The model related to the perception of job availability (i.e., "there are too few jobs available for me in my ideal commuting time") produces an estimated association consistent with our hypothesized causal relationship, in which increases in objective transit and active accessibility are associated with a higher probability of having a more positive perception of accessibility. In fact, this is the model in which the dependent variable is the most similar to the spatial availability of jobs definition, which refers directly to the number of jobs accessible, calculated using the impedance functions calibrated based on population behaviour. For this model, a one-standard-deviation increase in transit accessibility (about 0.07 jobs per transit user) is associated with 25% higher odds of being in a more positive response category, indicating that fewer respondents agree that there are too few jobs accessible within their ideal commuting time. Among non-active travellers, a one-standard-deviation increase in active accessibility (about 0.10 jobs per active traveller) is associated with 12% lower odds of being in a more positive response category. Similarly, the negative association of the active indicator also appears in the declined job offers model (-13% lower odds). For active travellers, on the other hand, the interaction with active accessibility is positive in both models (14% and 11%), but not statistically significant (p = 0.06 and p = 0.12), and the combined association for active travellers is close to one in both models (1.00 in the perception model and 0.97 in the declined job offers model). This means that the general negative association of active accessibility is attenuated, but not completely reversed, among people who walk or cycle (Figure 4). We believe that the negative association between active accessibility and responses among non-active travellers can be explained in terms of job competition, since increasing the number of jobs accessible for a mode that an individual does not use implies greater competition for jobs accessible through other modes, effectively reducing perceived accessibility for the mode a person actually uses. In addition, as previously shown in Figure 3, many of the areas with high active accessibility are located outside downtown cores, in more suburban areas with a low concentration of job opportunities. This reinforces the assumption that some suburban areas exhibit high active accessibility not because of greater job availability, but due to a very low number of active travellers,

Table 4: Weighted ordinal logistic regression models connecting objective to perceived accessibility and to employment outcome.

	Few jobs			Declined job offers		
Variable	**OR**	**SE**	**p-value**	**OR**	**SE**	**p-value**
Active job accessibility per active traveller	**0.88***	0.063	0.041	**0.87***	0.061	0.027
Active mode interaction: accessibility × user	1.14	0.068	0.062	1.11	0.066	0.12
Transit job accessibility per transit user	**1.25***	0.105	0.035	0.85	0.128	0.2
Transit mode interaction: accessibility × user	0.97	0.122	0.8	1.00	0.143	>0.9
Private vehicle accessibility per user	0.90	0.091	0.3	1.04	0.110	0.7
Number of Cars	0.94	0.106	0.6	1.23	0.108	0.051
Number of Bikes	1.11	0.059	0.084	0.97	0.054	0.6
Preferred mode match most frequent mode	1.14	0.158	0.4	1.41	0.187	0.067
Monthly or annual public transit pass	**0.67***	0.160	0.011	**0.59****	0.185	0.005
Public transit availability in the neighbourhood	0.75	0.260	0.3	1.07	0.186	0.7
Indigenous or racialized person	**0.66****	0.150	0.006	**0.50*****	0.161	<0.001
Temporary visa or refugee	1.43	0.239	0.14	0.95	0.314	0.9
Person with disability	0.68	0.197	0.050	**0.64***	0.177	0.013
Woman, transgender or non-binary person	1.06	0.137	0.7	1.00	0.150	>0.9
Age	1.01	0.006	0.060	1.01	0.006	0.052
Household size	1.06	0.049	0.3	0.97	0.055	0.5
Household income			0.13	***		<0.001
Less than $40,000	—	—		—	—	
$40,000 to $80,000	0.80	0.186		1.25	0.201	
$80,000 to $120,000	1.22	0.217		1.87	0.202	
More than $120,000	1.19	0.193		2.73	0.219	
Population density (people/km2)	**1.20***	0.070	0.010	**1.20****	0.056	0.001
Population centre class			0.4	***		<0.001
Rural	—	—		—	—	
Small	1.46	0.447		0.70	0.354	
Medium	2.27	0.457		3.51	0.379	
Large	1.35	0.311		1.18	0.264	
Residuals: Active accessibility	1.09	0.073	0.2	0.99	0.072	>0.9
Residuals: Transit accessibility	0.89	0.086	0.2	1.00	0.101	>0.9
Residuals: Private vehicle accessibility	**1.17***	0.065	0.014	1.06	0.076	0.4
No. Obs.	2,480			2,365		
Wald F	2.87			6.99		
p-value	<0.001			<0.001		
AIC	7,768			7,176		

Abbreviations: OR = Odds Ratio, SE = Standard Error

*p<0.05; **p<0.01; ***p<0.001.

Accessibility indicators, population density, and first-stage residuals are standardized; odds ratios correspond to a one-standard-deviation increase.

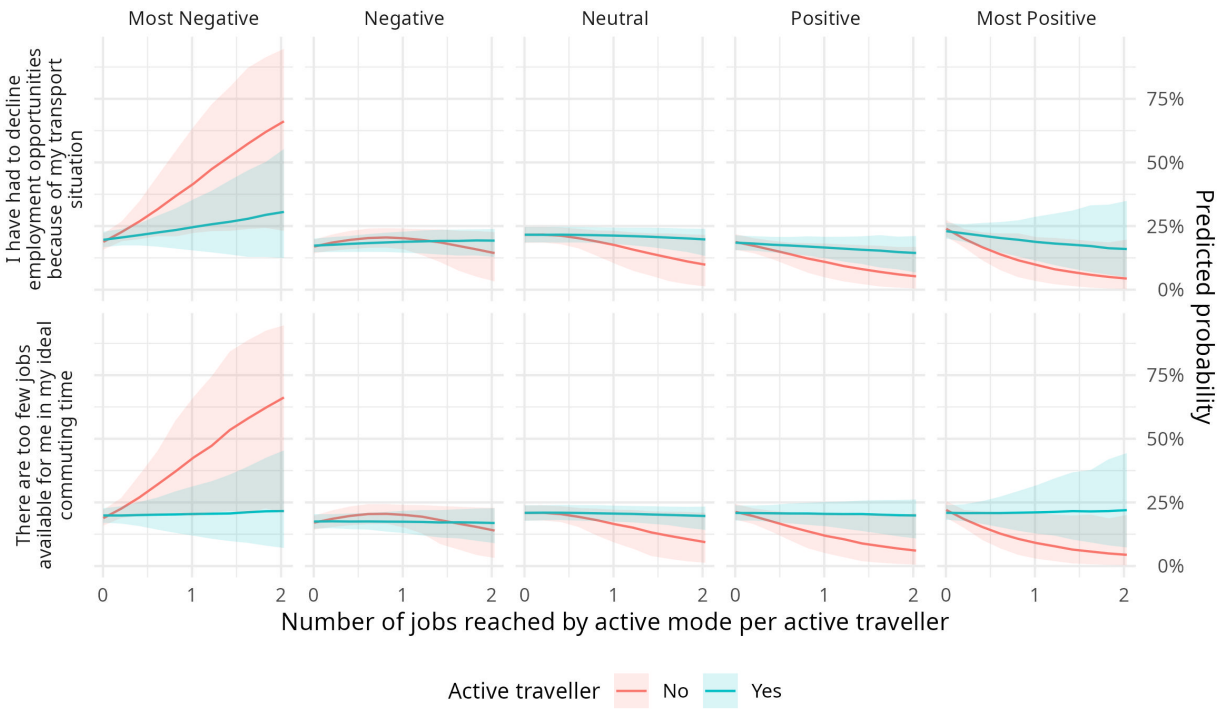


Figure 4: Average marginal effects for active accessibility between active and non-active travellers (with 95% confidence intervals).

resulting in a higher rate of jobs per person - reflecting low competition for jobs accessible by walking and cycling rather than higher active job availability.

Table 5 presents the control variable values applied for each individual profile, while Figure 5 displays the specific effect of increasing transit accessibility across the selected individual profiles for the "few jobs available" model. For all of them, we observe a decrease in negative perceptions as objective transit accessibility increases. For the transport-disadvantaged individual profile, the "Most negative" perception has a high probability at low levels of objective accessibility. In fact, at the median objective transit accessibility (0.12 jobs accessible by public transit per transit user, calculated based on all NSTP respondents), the two most negative perception categories ("Most negative" and "Negative") together account for over 54% of the total probability, resulting in a high likelihood that individuals in the transport-disadvantaged group will have negative perceptions at this accessibility level. For the most prevalent population, the initial pattern is already much more positive than that of the group at a transportation disadvantage, and the decline in negative perceptions also occurs more rapidly, with the combined probability of positive responses ("Positive" and "More positive") surpassing that of negative responses ("Negative" and "More negative") by approximately 0.10 jobs per public transit user - which is lower than the observed median. For the advantaged population, we can see that even at very low levels of accessibility, the positive perceptions already present the highest probabilities and increasing transit accessibility only reinforces these positive perceptions.

Regarding the control variables, in the model related to "there are too few jobs available for me in my ideal commuting time", having a monthly or annual public transit pass is associated with 33% lower odds of being in a more positive category. Being an Indigenous or racialized person is associated with a reduction of 34% in the odds of more positive perceptions. Regarding the population density, a one-standard-deviation increase (12,899 people/km^2) was associated with 20% higher odds for both models.

In the model related to "I have had to decline employment opportunities because of my transport

Table 5: Individual profiles to estimate probabilities.

Variable	Transport-disadvantaged	Most prevalent population	Advantaged group
Number of Cars	0	1	2
Number of Bikes	0	1	1
Preferred mode match most frequent mode	No	Yes	Yes
Indigenous or racialized person	**Yes**	**Yes**	**No**
Temporary visa or refugee	Yes	No	No
Person with disability	**Yes**	**No**	**No**
Woman, transgender or non-binary person	Yes	No	No
Age	37	40	43
Household size	3	3	3
Household income	**Less than $40,000**	**More than $120,000**	**More than $120,000**
Population centre class	**Small**	**Large**	**Large**
Public transit availability in the neighbourhood	Yes	Yes	Yes
Monthly or annual public transit pass	**Yes**	**No**	**No**

Bold variables are statistically significant (p-value < 0.05) for at least one of the models.

situation", having a monthly or annual public transit pass was associated with 41% lower odds of being in a more positive category. Being Indigenous or racialized and having a disability were associated with 50% and 36% lower odds, respectively. In terms of income, higher income is associated with more positive outcome (fewer reports of declined job offers), with households earning more than $120,000 CAD having almost three times higher odds of being in a positive category compared to those earning less than $40,000 CAD. In relation to the census population centre class, living in a medium-sized population centre is associated with substantially more positive outcomes (over three times higher odds), while living in a small centre is associated with 30% lower odds, and living in a large centre is associated with 18% higher odds - all this when compared to rural areas.

4.3. Equity standards and their impact on transportation equity

In this section, we identify public transit accessibility thresholds using the "there are too few jobs available for me in my ideal commuting time" model, since only this indicator was statistically significant and associated with more likelihood of positive responses. Figure 5 shows the conservative, intermediate, and progressive public transit thresholds calculated for each individual profile. For the advantaged profile, a level of 0 jobs per transit user is already enough to result in positive perceptions, which means that the current transit accessibility levels would be sufficient and no individuals would fall below the minimum accessibility if this profile was selected as the reference. However, it is worth noting that this result should not be interpreted as evidence that public transportation accessibility is irrelevant to the advantaged group's ability to reach employment. Instead, it demonstrates that the predicted probability of a positive perception for this profile is determined primarily by its other characteristics, such as higher household income and not being a racialized person, so that variation in accessibility has little additional effect. This is consistent with our DAG, which shows that individual resources mediate the relationship between objective accessibility, perceived accessibility, and employment outcomes, such that the more advantaged group has a greater ability to overcome low spatial accessibility through other means.

We did not calculate limits for active accessibility because, as previously mentioned, the identified association between active accessibility and active travellers for both models is close to zero and not statistically significant. Under these conditions, the predicted probabilities remain virtually constant even if we extrapolate the variation in accessibility values. This result presents the limitations of potential policies focused only on increasing active accessibility to promote a more positive perceived accessibility and employment outcomes: an approach that intervenes only in the spatial dimension of accessibility, without considering the capacities and travel preferences of the population, may have a limited effect on residents' perceived accessibility and employment outcome.

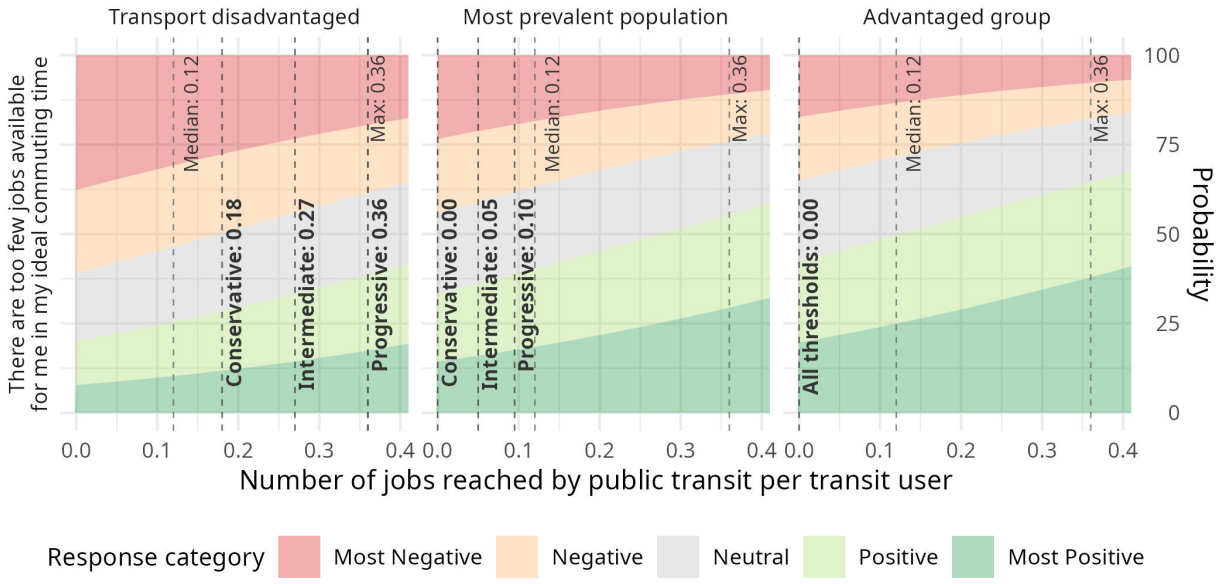


Figure 5: Public transit accessibility thresholds based on selected individual profiles.

These thresholds show how, depending on how the socioeconomic and transportation parameters are controlled and how the individual profiles are defined, different sufficient levels of accessibility are obtained. In fact, if one chooses to define thresholds based on the advantaged profile, it might lead to the conclusion that the current accessibility state is already sufficient and that no intervention or improvement is required. Alternatively, transport-disadvantaged groups might still hold negative perceptions depending on whether the threshold based on the most prevalent population is selected. Therefore, the only way to ensure that transport-disadvantaged groups have a better chance of experiencing positive perceptions and employment outcomes is to define minimum accessibility levels based on the controls tailored to these populations, setting socioeconomic and transportation parameters that represent them, thereby adopting a truly equity-based standard.

Considering the feasibility for policy interventions, we suggest the transit thresholds defined in terms of the transport-disadvantaged profile as examples of equity standards. Around 72% of the study area population lives in areas with transit accessibility levels lower than the conservative equity standard (0.18), while 93% live in areas with transit accessibility levels lower than 0.27 jobs accessible by transit per transit user (intermediate equity standard), and almost the entire study area population lives in areas below the progressive equity standard (0.36, which is also the maximum transit accessibility observed in the study area). Regarding the low-income population, 62% live in areas with an accessibility gap under the conservative standard, 89% under the intermediate standard, and almost all live in areas with an accessibility gap under the most progressive standard.

In terms of the severity of transport disadvantage (FGT-1), the average gap among deprived individuals amounts to 40% under the conservative standard, meaning that those who do not meet the threshold fall short of it by roughly one-third on average (about 0.07). For the low-income population, the average shortfall is around 30% (about 0.05). For the intermediate standard, the average shortfall for the general population is 54% (about 0.15), while it is 46% (0.12) for the low-income group. For the progressive equity standard, the average shortfall for the general population is about 0.23 (65% of the standard), and about 0.21 (58%) for the low-income group. These FGT measures show that, as the minimum transit accessibility levels become more demanding when moving from conservative to progressive standards, not only do more individuals fall below the threshold, but those in accessibility gap are, on average, substantially further from

achieving it. They also show that the low-income population is, in general, located in areas with greater public transportation accessibility than the general population, meaning that low-income individuals who fall below a certain threshold are, on average, closer to reaching it and would need comparatively smaller improvements in accessibility to reach it.

Figure 6 shows the public transit accessibility gap in each CT, calculated as the difference between the CT's transit accessibility level and each equity standard. All census subdivisions contain CTs with a transit accessibility gap, regardless of the type of standard. For the conservative standard, only 29% of the CTs (428 out of the 1,489 CTs in the study area) present transit accessibility levels above the standard, with most of them (87%) located in the City of Toronto. This share of CTs without an accessibility gap decreases to less than 8% for the intermediate equity standard (110 CTs). Only one CT, located in Toronto, does not present an accessibility gap when evaluated under the progressive standard. This unique CT corresponds to the area surrounding the St. George Subway Station in Downtown Toronto, connected to the two subway lines that cross the city and one of which extends into neighbouring municipalities, running from north to south and from east to west. This CT is bounded by Spadina Avenue to the west and Bloor Street to the south, with many connections to the public transit system, including a streetcar line on Spadina.

In terms of absolute population living in accessibility insufficiency when compared to the conservative standard, the City of Toronto has the largest population in this situation (almost 997,000 people, 36% of its residents), followed by Mississauga (over 650,000 people, 90%), Brampton (almost 649,000, 99%), and Hamilton (almost 570,000, 100%). For the low-income population, Toronto again ranks first (almost 121,000 individuals, 33% of its low-income residents), followed by Hamilton (60,260, 100%), and Mississauga (over 57,500, 86%).

We believe that the areas with accessibility gaps identified here can help guide future investments in public transportation services and infrastructure. Although suburban areas such as Vaughan and Markham exhibit large absolute accessibility gaps, viewing these differences with a vertical equity perspective suggests different priorities. If focus is on the absolute number and proportion of low-income residents living in areas with an accessibility gap, Hamilton and Toronto are candidates for intervention. Toronto has the largest low-income population below the standard, while in Hamilton all low-income residents fall below it.

Because objective accessibility depends on transport and land-use factors, gaps can be reduced through two broad types of intervention. First, improvements in transit service aiming to increase public transit accessibility, such as higher service frequencies, new routes, or rapid transit lines, reduce travel times and expand the number of jobs reachable within typical commuting times. Second, land-use policies that bring employment closer to transit, such as transit-oriented development and mixed-use zoning, can increase the number of opportunities reachable without changes to the network.

For instance, the City of Hamilton is currently implementing its first Light Rail Transit (LRT) line in Downtown Hamilton, connecting the eastern and western parts of the city while also crossing Downtown, an area that concentrates a large share of the city's available jobs. Provided the city mitigates potential displacement and relocation of its low-income population resulting from land price increases associated with the LRT, its inclusion is likely to have a significant impact on transportation equity, since the LRT crosses the areas with accessibility gaps and the largest concentration of low-income residents in the city.

5. Critical analysis and limitations

In this study, we present a model-based approach to defining equity standards that achieve a desired normative outcome, based on the estimated relationship between objective accessibility, perceived accessibility, and an employment outcome. We consider this approach our main contribution to transportation research, and more specifically to the literature on the equitable distribution of transportation benefits. Because the resulting standards are based on how residents actually experience accessibility, they are also easier to communicate to policy practitioners and the general public.

Defining minimum levels of accessibility necessarily requires selecting a threshold at which objective accessibility becomes "sufficient" (Martens, 2016b). Following Pereira and Karner (2021), we combine egalitarian and sufficientarian perspectives of distributive justice and define this point as the minimum level

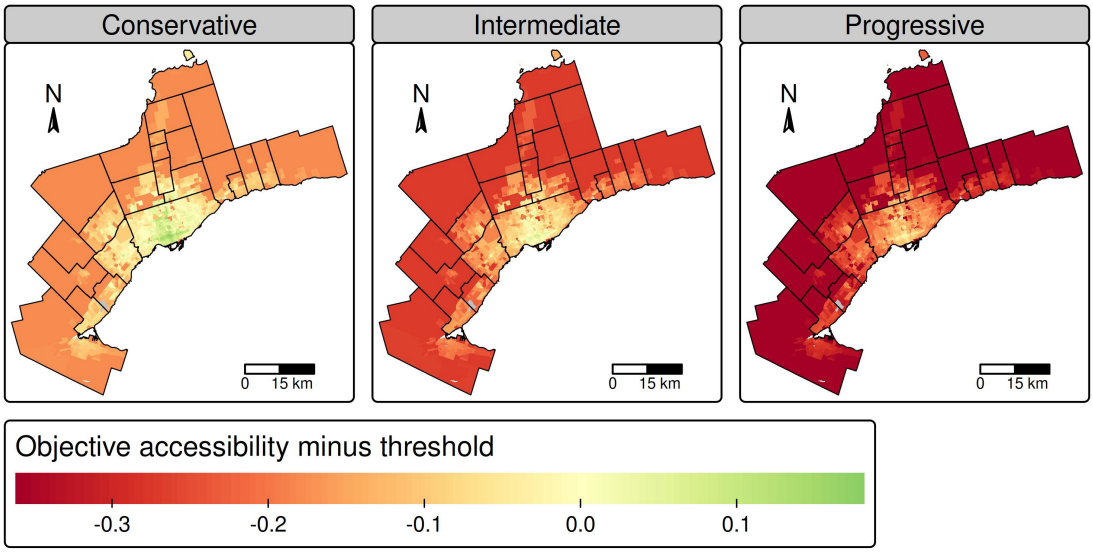


Figure 6: Difference between objective accessibility and thresholds.

of accessibility (sufficientarianism) at which a transport-disadvantaged profile (egalitarianism) is more likely to give a non-negative response than a negative one (conservative standard), or a positive response than a negative one (progressive standard). In our view, it is important to make explicit the moral foundations that support this choice, because different theories of justice lead to different perspectives on how policies should manage the distribution of transport-related benefits.

However, these standards could have been defined in many other ways, both in terms of the theory of justice guiding their definition and in terms of the threshold considered sufficient. This choice is ultimately normative, and it should not rest solely with researchers. As stated by Karner et al. (2020), transport equity is only one step towards achieving transport justice: a condition in which no person or group is disadvantaged by a lack of access to the opportunities they need to lead a dignified and healthy life. Achieving this condition requires transforming the structures and processes that lead to the unequal distribution of the multiple externalities of transport across population groups and space.

Consequently, the minimum levels of accessibility we propose are intended to guide transportation planning towards a more equitable distribution of transportation benefits, as an initial or an intermediate stage to be followed by a participatory and democratic process to collectively select the most appropriate standard. Residents and other stakeholders must be able to actively participate in, and influence, the decisions that affect their lives. Such a process could define not only the final standard but also involve the public throughout the entire modelling process. This could include deciding which variables should be included in the models, which profile should serve as the reference for defining thresholds, and how the thresholds themselves should be defined. Although we consider this the most appropriate way to define standards, it is beyond the scope of this study to develop democratic plenaries to discuss the suggested standards.

In addition, the scope of our standards is limited, and we do not propose them as fixed or universal benchmarks applicable to other regions. By definition, objective accessibility depends on the study area's land-use and transport characteristics and on the measure selected to represent it, and we would not expect or intend for our thresholds to be transferred elsewhere. However, while our results are context-dependent, we believe that our modelling approach is not.

In terms of the opportunity type, we selected job accessibility as our indicator to define equity standards for a few reasons. First, employment accessibility is widely recognized as a key indicator of how well a city or region serves its residents (Bania et al., 2008; Owen and Levinson, 2015; Shen, 1998), and as an

indicator of the "quality of urban living" (Wachs and Kumagai, 1973; Handy, 2020), since employment provides the financial ability to support other life activities (Merlin and Hu, 2017). Second, calculating accessibility requires the use of impedance functions, and we decided to use calibrated functions based on empirical data to better represent people's travel behaviour. Accessibility measures also require land-use data on both population (for competitive measures) and the number of opportunities. Both requirements are met by Census microdata specifically for employment and the labour force, which is not the case for other types of opportunity. Third, the NSTP survey includes questions on perceived job accessibility and employment outcomes specifically, which directly enabled our approach of linking objective job accessibility to individuals' perceptions and outcomes related to this opportunity type. We believe, however, that our methodology can be extended to other opportunity types essential for a meaningful life, such as healthcare and education, provided that the analyst has the necessary data on hand to perform the analysis.

Although the NSTP provides an unprecedented snapshot of transport poverty across Canada, the survey does not contain a question in which the respondent answers objectively whether they are satisfied or not with their mobility experiences or overall transport conditions. More specifically, the respondents use a continuum scale to answer their level of agreement with specific statements. The absence of a categorical or binary variable referring to satisfaction level led us to categorize the outcome, in order to identify which response category the respondent belongs to, varying from most negative to most positive. In fact, with the information we have, we cannot infer if the person is satisfied or not, but rather whether they are more or less satisfied in comparison to others. The underlying rationale for our standards is therefore relative: they identify the minimum objective accessibility at which the most disadvantaged profile is no longer more likely than not to be among the respondents reporting the most negative perceptions in the region.

It is worth mentioning that other outcome categorizations or numbers of categories are possible. For instance, one might decide to convert the continuum scale to a Likert scale based on fixed intervals, classifying those with agreement levels from 0 to 20%, 20 to 40%, and so on, in pre-determined intervals. The problem with the fixed-interval approach is that the analyst is normatively defining levels of satisfaction for respondents that might not reflect their actual feelings. Because of this, we decided to categorize responses based on the perception scale distribution and use the sample population to evaluate those who are more and less satisfied.

To test the sensitivity of our results to the choice of categorization type, we estimated a new perception model using a fixed-interval categorization that does not depend on the sample's distribution, instead of a quintile-based categorization (classifying responses from 0 to 20 as most positive, 20 to 40 as positive, 40 to 60 as neutral, 60 to 80 as negative, and 80 to 100 as most negative). The results of this alternative model (Appendix A.4) are consistent with those of the model created using the quintile classification. The transit accessibility, on which the proposed standards are based, retain its sign, magnitude, and significance ($OR = 1.28$, $p = 0.021$, compared to $OR = 1.25$, $p = 0.035$ in our main specification) and so does the first-stage residual of private vehicle accessibility ($OR = 1.18$, $p = 0.008$). Overall, the differences suggest changes in precision rather than in direction. However, the quintile-based categorization caused, in general, the control variables to lose part of their predictive power when compared to linear regression models created using the previous continuous forms. As shown in Table A.3, in general, and similarly to the categorical models, the accessibility estimates are consistent with our hypothesized causal effect. In contrast, the control variables presented a better performance in explaining the outcomes in the linear regressions, such as in the case of the number of bikes and the "temporary visa or refugee" variable, which did not present statistical significance in the categorical version of perception model.

An additional limitation concerns the interpretation of our declined job offers indicator. Agreeing or disagreeing with this question presupposes that the respondent had looked for a job and subsequently received a job offer that they declined. In this sense, individuals who did not look for work (including those who may have given up searching because they anticipated that available jobs would be inaccessible due to their transportation situation) would not be recorded as having declined an offer, regardless of the extent to which transportation constraints limit their employment prospects. This mechanism is conceptually related to the discussion of adaptive preferences: just as some people may adjust downward their perceptions of job scarcity in response to low accessibility, others may give up on the job search completely. Consequently, the indicator of declined job offers likely captures transportation-related employment disadvantage only among

those who remain active in the labour market and may underestimate the prevalence of transportation disadvantage among those most severely excluded from employment opportunities. We cannot assess this selection directly using the current cross-sectional data from the NSTP, which do not allow us to distinguish those who are not looking for work from those who are actively seeking employment but have not received offers to decline.

Regarding the relationship between perceived and objective accessibility, the transit accessibility indicator was the most consistent with our hypothesized causal effect in the perception-based model, through which increasing accessibility was associated with more positive perceptions. For active accessibility, the general association was negative and close to null for active travellers in both models, so we could not derive active thresholds for any profile. These results led us to propose equity standards, exclusively, based on the perception-based model and for public transit mode.

Regarding evidence of endogeneity, only accessibility by private vehicle was identified as endogenous, and only in the perception-based model, as demonstraded by the significant first-stage residual in the ordinal model and by the Wu-Hausman test in its linear version. This result reinforces our view that one cannot assume a priori that accessibility indicators are exogenous and that the instrumental variable framework is necessary, at least for the perception model, from which the proposed standards are derived. However, as is generally the case in instrumental variable designs (Cunningham, 2021; Huntington-Klein, 2021), our assumption that the instruments affect perceived accessibility and employment outcome only through objective accessibility cannot be directly tested. Even so, adopting an instrumental variable approach was the most suitable alternative for accounting for potential endogeneity given the use of cross-sectional surveys.

Our instrumental variable strategy was not specifically designed to address residential self-selection, and although it offers a degree of protection against confounding factors more broadly, we cannot exclude the possibility that unobserved commuting preferences and constraints that determine residential location choices are correlated with our spatial filter instruments. A more definitive test would require longitudinal data tracking individuals across their residential moves over the years, or an instrument based on a source of variation known to be exogenous to residential choice; neither of these options is available in the current cross-sectional NSTP data, nor in the Canadian Census.

For future research, we suggest exploring additional objective accessibility measures to further evaluate the relationship between objective and perceived accessibility. For instance, the raw multimodal spatial availability indicator could be tested. This indicator reflects the number of jobs accessible by each transport mode, without standardizing by the number of people competing for those jobs. We decided to use multimodal spatial availability per person because we believe that competition matters when evaluating perceptions related to employment outcomes, as demonstrated by previous studies (Jin and Paulsen, 2018; Merlin and Hu, 2017; Miller, 2005); and because the standardized version allows for comparisons across cities with different labour force and job opportunity sizes.

Regarding the spatial unit used, the CTs, we recognize that this unit is not the most detailed among census geographies, being less detailed than dissemination areas or dissemination blocks. We decided to use CTs due to the large size of the study area, in order to provide a regional analysis rather than focus on only one city, considering this entire basin to calculate the accessibility indicators and select the sample. This choice is motivated by the recognition of commuting flows across municipalities, a pattern that is itself one of the parameters used by Statistics Canada when defining Census Metropolitan Areas. Another limiting factor that led us to adopt the CT level was that we could not estimate travel times at the dissemination area level for the entire study area due to the memory limitations of our computational resources. Despite that, the initial tests performed with a smaller study area (for the City of Toronto) did not show differences in terms of the sign and statistical significance of the accessibility coefficients between regressions - probably because the indicators are standardized based on the labour force size.

6. Conclusions

The main objective of this study was to propose a model-based methodology to define equity standards for transportation planning, based on the relationship between objective job accessibility, people's perceived

accessibility, and an employment outcome. The study area covers municipalities around the cities of Toronto and Hamilton, in southern Ontario, Canada. To this end, we used data from the 2021 Census of Population and the recently released National Survey on Transport Poverty (Tiznado-Aitken et al., 2025), combining (place-based) objective accessibility measures with variables at the individual level.

Our approach is distinguished from other studies by translating estimated associations between objective accessibility, perceived accessibility, and an employment outcome into planning thresholds and by proposing thresholds based on objective accessibility calculated from spatial land-use and transport data and validating them against how accessibility is actually experienced by residents. Although the proposed standards are presented at the neighbourhood level, they indirectly also incorporate individual-level characteristics, since these standards are obtained from individuals' self-reported perceptions and outcomes in the NSTP. In this way, these thresholds can assist policymakers in interpreting how different levels of objective accessibility are associated with residents' perceived ability to reach employment opportunities under existing social and spatial constraints. This increases the potential for government adoption and enhances the comprehensibility and relevance of such standards for community members and those experiencing unequal accessibility. Furthermore, identifying sufficient job accessibility levels can help assess and highlight areas at higher risk of transport poverty, particularly those with a higher presence of transport-disadvantaged populations.

Objective accessibility models were created using multimodal spatial availability per person, considering walking and cycling (later combined as active transportation), public transit, and private vehicles.

Both models were statistically significant overall (Wald F-tests, $p < 0.001$). In the perceived accessibility model, the public transit accessibility estimates are consistent with our hypothesized relationship, with a one-standard-deviation increase in transit accessibility (about 0.09 jobs per transit user) associated with 25% higher odds of being in a more positive response category ($p < 0.05$). For active travellers, the combined association is close to null in both models.

Using the perception-based model and for the public transit mode, we suggest equity standards based on the accessibility threshold that results in a higher likelihood of positive perceptions for people from transport-disadvantaged groups. We suggest three equity standards: a conservative standard (0.18 jobs reached by public transit per transit user), an intermediate standard (0.27), and a progressive standard (0.36).

Our results show that most of the study area falls short of the proposed equity standards, regardless of their category (conservative, intermediate, or progressive). About 72% of the study area population and 62% of the low-income population live in areas with transit accessibility below the conservative standard, while 92% of the general population and 88% of the low-income population live in areas below the intermediate standard, and almost the entire study area population lives below the progressive standard.

Finally, we believe our study contributes to the ongoing discussion of equity in transportation, and that equity standards based on, or inspired by, our methodology can serve as a model-based starting point for debate among residents and other stakeholders within a participatory and democratic process to collectively define the most appropriate standard.

CRediT authorship contribution statement

(include after the review)

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Open science statement

To facilitate collaboration and further analysis and to ensure the transparency and reproducibility of our study, we developed this paper using literate programming, with the data analysis code accessible

through our GitHub page (`https://github.com/dias-bruno/NSTP/`), in alignment with best practices in open spatial data science (Páez, 2021; Arribas-Bel et al., 2021).

Declaration of competing interest

The authors declare no conflicts of interest.

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Appendix

Table A.1: Impedance functions calibrated for the census subdivisions of the study area.

Census Division	Census Code	Mode	Impedance function	Parameter 1*	Parameter 2*	AIC
Durham	3518	Car-motor	Gamma	1.92	0.07	1714717
	3518	Walk	Exponential	0.09		42507
	3518	Transit	Gamma	3.33	0.06	121000
	3518	Bike	Gamma	1.89	0.08	3829
York	3519	Car-motor	Gamma	2.16	0.08	2549528
	3519	Transit	Normal	54.89	28.44	200698
	3519	Walk	Exponential	0.09		58149
	3519	Bike	Lognormal	2.75	0.81	8060
Toronto	3520	Transit	Gamma	3.60	0.08	1880937
	3520	Car-motor	Gamma	2.33	0.09	3942809
	3520	Walk	Gamma	1.33	0.08	449370
	3520	Bike	Lognormal	2.94	0.65	123658
Peel	3521	Car-motor	Gamma	2.33	0.09	3351993
	3521	Walk	Exponential	0.08		72360
	3521	Bike	Lognormal	2.87	0.73	8853
	3521	Transit	Gamma	3.28	0.06	468686
Halton	3524	Car-motor	Gamma	1.92	0.07	1264978
	3524	Transit	Normal	53.20	26.65	75851
	3524	Walk	Lognormal	1.90	1.08	40360
	3524	Bike	Gamma	1.99	0.11	6229
Hamilton	3525	Car-motor	Gamma	1.85	0.07	1323879
	3525	Walk	Exponential	0.08		61816
	3525	Transit	Gamma	2.24	0.05	128266
	3525	Bike	Lognormal	2.83	0.75	9830

Note:
For each probability distribution, Parameter 1 and Parameter 2 correspond to their standard defining parameters: the mean and standard deviation (for lognormal and normal), the rate and shape (for gamma), and the power (for exponential). The AIC represents the Akaike Information Criterion used to assess model fit.

Table A.2: Instrument relevance (weak-instrument F-tests) and endogeneity (Wu–Hausman) diagnostics for linear (2SLS) models estimated on the original continuous dependent variables.

	Few jobs		Declined job offers	
Diagnostic test	F-statistic	p-value	F-statistic	p-value
Weak instruments: active accessibility	83.42***	3.41e-80	105.21***	5.01e-99
Weak instruments: transit accessibility	432.25***	1.06e-320	389.90***	3.43e-294
Weak instruments: private vehicle accessibility	586.06***	0	608.90***	0
Weak instruments: active × user	27.02***	1.39e-26	19.65***	3.64e-19
Weak instruments: transit × user	34.48***	5.54e-34	27.95***	1.8e-27
Wu–Hausman (endogeneity test)	2.32*	0.0407	1.14	0.338

*p<0.05; **p<0.01; ***p<0.001

References

Akaike, H., 1974. A new look at the statistical model identification. IEEE Transactions on Automatic Control 19, 716–723. URL: https://ieeexplore.ieee.org/document/1100705, doi:10.1109/TAC.1974.1100705. conference Name: IEEE Transactions on Automatic Control.

Allen, J., Farber, S., 2019. Sizing up transport poverty: A national scale accounting of low-income households suffering from inaccessibility in canada, and what to do about it. Transport Policy 74, 214–223. URL: https://www.sciencedirect.com/science/article/pii/S0967070X18304736, doi:10.1016/j.tranpol.2018.11.018.

Antonakis, J., Bendahan, S., Jacquart, P., Lalive, R., 2010. On making causal claims: A review and recommendations. The Leadership Quarterly 21, 1086–1120. URL: https://linkinghub.elsevier.com/retrieve/pii/S1048984310001475, doi:10.1016/j.leaqua.2010.10.010.

Arribas-Bel, D., Green, M., Rowe, F., Singleton, A., 2021. Open data products-a framework for creating valuable analysis ready data. Journal of Geographical Systems 23, 497–514. URL: https://doi.org/10.1007/s10109-021-00363-5, doi:10.1007/s10109-021-00363-5.

Awaworyi Churchill, S., 2020. Ethnic diversity and transport poverty. Transportation Research Part A: Policy and Practice 139, 297–309. URL: https://www.sciencedirect.com/science/article/pii/S0965856420306698, doi:10.1016/j.tra.2020.07.012.

Baber, H., 2007. Adaptive preference. Social Theory and Practice 33, 105–126. URL: http://www.jstor.org/stable/23558512.

Bania, N., Leete, L., Coulton, C., 2008. Job access, employment and earnings: Outcomes for welfare leavers in a us urban labour market. Urban Studies 45, 2179–2202. URL: https://doi.org/10.1177/0042098008095864, doi:10.1177/0042098008095864.

Bastiaanssen, J., Johnson, D., Lucas, K., 2021. Does better job accessibility help people gain employment? the role of public transport in great britain. Urban Studies 59. URL: https://journals.sagepub.com/doi/10.1177/00420980211012635, doi:10.1177/00420980211012635.

Bastiaanssen, J., Johnson, D., Lucas, K., 2025. The relationship between individual employment probabilities and accessibility to matching jobs: A study of the netherlands. Transportation Research Part A: Policy and Practice 193, 104398. URL: https://www.sciencedirect.com/science/article/pii/S0965856425000266, doi:10.1016/j.tra.2025.104398.

Behrendt, F., Sheller, M., 2024. Mobility data justice. Mobilities 19, 151–169. URL: https://doi.org/10.1080/17450101.2023.2200148, doi:10.1080/17450101.2023.2200148.

Boisjoly, G., El-Geneidy, A.M., 2017. How to get there? a critical assessment of accessibility objectives and indicators in metropolitan transportation plans. Transport Policy 55, 38–50. URL: https://www.sciencedirect.com/science/article/pii/S0967070X1630436X, doi:10.1016/j.tranpol.2016.12.011.

Canada, S., 2023. 2021 Census of Population. Technical Report. Ottawa. URL: https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/about-apropos/about-apropos.cfm?Lang=E#aa1. last Modified: 2022-11-30.

Canada, S., 2025. Canadian Public Transit Network Database. Technical Report. URL: https://www150.statcan.gc.ca/n1/pub/23-26-0003/232600032025001-eng.htm.

Cao, X., Mokhtarian, P.L., Handy, S.L., 2009. Examining the impacts of residential self-selection on travel behaviour: A focus on empirical findings. Transport Reviews 29, 359–395. URL: https://doi.org/10.1080/01441640802539195, doi:10.1080/01441640802539195.

Church, R.L., Marston, J.R., 2003. Measuring accessibility for people with a disability. Geographical Analysis 35, 83–96.

Cunningham, S., 2021. Causal inference: The mixtape. Yale University Press, New Haven.

De Vos, J., Pot, F., van Lierop, D., El-Geneidy, A., 2025. Perceived accessibility: An introduction. Transportation Research Part A: Policy and Practice 199, 104553. URL: https://www.sciencedirect.com/science/article/pii/S0965856425001818, doi:10.1016/j.tra.2025.104553.

Delignette-Muller, M.L., Dutang, C., 2015. fitdistrplus: An r package for fitting distributions. Journal of statistical software 64, 1–34.

Delmelle, E., Nilsson, I., Adu, P., 2021. Poverty suburbanization, job accessibility, and employment outcomes. Social Inclusion 9, 166–178. URL: https://www.cogitatiopress.com/socialinclusion/article/view/3735, doi:10.17645/si.v9i2.3735.

Table A.3: Weighted linear regression models connecting objective to perceived accessibility and to employment outcome.

	Few jobs			Declined job offers		
Variable	**Beta**	**SE**	**p-value**	**Beta**	**SE**	**p-value**
(Intercept)	**69*** **	6.92	<0.001	**53*** **	7.08	<0.001
Predicted active accessibility per user	**10** **	3.34	0.002	**13***	5.73	0.024
Predicted transit accessibility per user	**-59** **	18.8	0.002	17	22.3	0.4
Predicted private vehicle accessibility per user	7.7	4.77	0.11	-2.1	6.19	0.7
Number of Cars	0.63	1.35	0.6	-0.62	1.32	0.6
Number of Bikes	**-2.5** **	0.802	0.002	0.93	0.655	0.2
Preferred mode match most frequent mode	-1.1	2.10	0.6	**-5.4***	2.22	0.014
Monthly or annual public transit pass	**5.4** **	2.07	0.010	**10*** **	2.54	<0.001
Public transit availability in the neighbourhood	**12*** **	3.21	<0.001	-1.9	2.62	0.5
Indigenous or racialized person	3.7	1.94	0.057	**18*** **	2.10	<0.001
Temporary visa or refugee	**-7.9***	3.39	0.020	6.4	3.79	0.093
Person with disability	4.5	2.37	0.056	3.3	2.07	0.11
Woman, transgender or non-binary person	-0.51	1.77	0.8	0.19	1.87	>0.9
Age	-0.11	0.069	0.11	**-0.16***	0.071	0.024
Household size	-0.75	0.648	0.2	0.53	0.713	0.5
Household income			0.11	***		<0.001
Less than $40,000	—	—		—	—	
$40,000 to $80,000	0.82	2.30		-5.8	2.50	
$80,000 to $120,000	-3.0	2.69		-14	2.76	
More than $120,000	-4.8	2.54		-27	2.81	
Population density (people/km2)	**0.00*** **	0.000	<0.001	**0.00** **	0.000	0.007
Population centre class	*		0.043	***		<0.001
Rural	—	—		—	—	
Small	-12	6.23		9.5	5.56	
Medium	-15	5.97		-17	4.35	
Large	-5.1	3.65		-5.6	3.75	
Active mode interaction: accessibility × user	**-15** **	5.58	0.006	5.7	12.4	0.6
Transit mode interaction: accessibility × user	5.8	15.0	0.7	**57*** **	16.8	<0.001

Abbreviation: SE = Standard Error
*p<0.05; **p<0.01; ***p<0.001

Table A.4: Alternative weighted ordinal logistic regression model connecting objective to perceived accessibility using a fixed-interval categorization for the dependent variable.

Variable	**OR**	**SE**	**p-value**
Active job accessibility per active traveller	0.89	0.062	0.058
Active mode interaction: accessibility × user	1.08	0.070	0.3
Transit job accessibility per transit user	**1.28***	0.109	0.021
Transit mode interaction: accessibility × user	0.97	0.125	0.8
Private vehicle accessibility per user	0.90	0.092	0.2
Number of Cars	0.93	0.098	0.4
Number of Bikes	1.09	0.055	0.11
Preferred mode match most frequent mode	1.10	0.164	0.6
Monthly or annual public transit pass	**0.70***	0.155	0.021
Public transit availability in the neighbourhood	0.78	0.228	0.3
Indigenous or racialized person	**0.68***	0.158	0.015
Temporary visa or refugee	1.34	0.231	0.2
Person with disability	0.73	0.187	0.10
Woman, transgender or non-binary person	0.97	0.140	0.8
Age	**1.01***	0.006	0.027
Household size	1.09	0.050	0.084
Household income			0.2
Less than $40,000	—	—	
$40,000 to $80,000	0.86	0.179	
$80,000 to $120,000	1.28	0.214	
More than $120,000	1.21	0.195	
Population density (people/km2)	**1.20***	0.073	0.011
Population centre class			0.4
Rural	—	—	
Small	1.34	0.543	
Medium	2.10	0.452	
Large	1.24	0.291	
Residuals: Active accessibility	1.10	0.074	0.2
Residuals: Transit accessibility	0.90	0.101	0.3
Residuals: Private vehicle accessibility	**1.18****	0.064	0.008
No. Obs.	2,480		
Wald F	2.92		
p-value	<0.001		
AIC	7,480		

Abbreviations: OR = Odds Ratio, SE = Standard Error
*p<0.05; **p<0.01; ***p<0.001. Accessibility indicators, population density, and first-stage residuals are standardized; odds ratios correspond to a one-standard-deviation increase.

Dos Santos, B., Paez, A., 2026. Commuteca: Datasets and methods for job accessibility in canada. URL: https://github.com/dias-bruno/CommuteCA. r package version 5.0.

Dos Santos, B.D., Hanson, T., Páez, A., . Dignity in transportation: A scoping review and theoretical framework .

Duarte, L.B., Neto, R.d.M.S., Silva, D.F.C.d., 2023. The relevance of job accessibility to labour market outcomes: Evidence for the são paulo metropolitan region. Urban Studies 60. doi:https://doi.org/10.1177/00420980231165481.

El-Geneidy, A., Levinson, D., 2022. Making accessibility work in practice. Transport Reviews 42, 129–133. URL: https://doi.org/10.1080/01441647.2021.1975954, doi:10.1080/01441647.2021.1975954. _eprint: https://doi.org/10.1080/01441647.2021.1975954.

Foster, J., Greer, J., Thorbecke, E., 2010. The foster–greer–thorbecke (fgt) poverty measures: 25 years later. The Journal of Economic Inequality 8, 491–524.

Getis, A., Griffith, D.A., 2002. Comparative spatial filtering in regression analysis. Geographical Analysis 34, 130–140.

Geurs, K.T., van Wee, B., 2004. Accessibility evaluation of land-use and transport strategies: review and research directions. Journal of Transport Geography 12, 127–140. URL: https://www.sciencedirect.com/science/article/pii/S0966692303000607, doi:10.1016/j.jtrangeo.2003.10.005.

Griffith, D.A., 2003. Spatial Autocorrelation and Spatial Filtering: Gaining Understanding Through Theory and Scientific Visualization. Springer-Verlag, Berlin.

Handy, S., 2020. Is accessibility an idea whose time has finally come? Transportation Research Part D: Transport and Environment 83, 102319. URL: https://www.sciencedirect.com/science/article/pii/S1361920919315925, doi:10.1016/j.trd.2020.102319.

Handy, S.L., Clifton, K.J., 2001. Evaluating neighborhood accessibility: Possibilities and practicalities. Journal of transportation and statistics 4, 67–78.

Hansen, W.G., 1959. How accessibility shapes land use. Journal of the American Institute of Planners 25, 73–76. URL: https://doi.org/10.1080/01944365908978307, doi:10.1080/01944365908978307. _eprint: https://doi.org/10.1080/01944365908978307.

Hu, L., 2017. Job accessibility and employment outcomes: which income groups benefit the most? Transportation (Dordrecht) 44, 1421–1443.

Huntington-Klein, N., 2021. The effect: An introduction to research design and causality. 1st edition ed., Chapman and Hall/CRC, New York.

Jin, J., Paulsen, K., 2018. Does accessibility matter? understanding the effect of job accessibility on labour market outcomes. Urban Studies 55, 91–115. URL: https://doi.org/10.1177/0042098016684099, doi:10.1177/0042098016684099.

Johnson, D., Ercolani, M., Mackie, P., 2017. Econometric analysis of the link between public transport accessibility and employment. Transport Policy 60, 1–9. URL: https://www.sciencedirect.com/science/article/pii/S0967070X16303353, doi:10.1016/j.tranpol.2017.08.001.

Kain, J.F., 1968. Housing segregation, negro employment, and metropolitan decentralization. The Quarterly Journal of Economics 82, 175–197. URL: https://www.jstor.org/stable/1885893, doi:10.2307/1885893.

Karner, A., London, J., Rowangould, D., Manaugh, K., 2020. From transportation equity to transportation justice: Within, through, and beyond the state. Journal of Planning Literature 35, 440–459. URL: https://doi.org/10.1177/0885412220927691, doi:10.1177/0885412220927691.

Karner, A., Pereira, R.H.M., Farber, S., 2024. Advances and pitfalls in measuring transportation equity. Transportation URL: https://doi.org/10.1007/s11116-023-10460-7, doi:10.1007/s11116-023-10460-7.

Kim, S.O., Palm, M., Han, S., Klein, N.J., 2024. Facing a time crunch: Time poverty and travel behaviour in canada. Transportation Research Part D: Transport and Environment 126, 104028. URL: https://www.sciencedirect.com/science/article/pii/S136192092300425X, doi:10.1016/j.trd.2023.104028.

Le Gallo, J., Páez, A., 2013. Using synthetic variables in instrumental variable estimation of spatial series models. Environment and Planning A: Economy and Space 45, 2227–2242. URL: https://journals.sagepub.com/doi/10.1068/a45443, doi:10.1068/a45443.

Liddell, T.M., Kruschke, J.K., 2018. Analyzing ordinal data with metric models: What could possibly go wrong? Journal of Experimental Social Psychology 79, 328–348. URL: https://www.sciencedirect.com/science/article/pii/S0022103117307746, doi:10.1016/j.jesp.2018.08.009.

Litman, T., 2024. Evaluating Transportation Equity: Guidance for Incorporating Distributional Impacts in Transport Planning. Technical Report. URL: https://www.vtpi.org/equity.pdf.

Lucas, K., 2012. Transport and social exclusion: Where are we now? Transport Policy 20, 105–113. URL: https://www.sciencedirect.com/science/article/pii/S0967070X12000145, doi:10.1016/j.tranpol.2012.01.013.

Lucas, K., 2019. A new evolution for transport-related social exclusion research? Journal of Transport Geography 81, 102529. URL: https://www.sciencedirect.com/science/article/pii/S0966692319303552, doi:10.1016/j.jtrangeo.2019.102529.

Lumley, T., 2011. Complex surveys: a guide to analysis using R. John Wiley & Sons.

Luz, G., Portugal, L.d.S., 2022. Understanding transport-related social exclusion through the lens of capabilities approach. Transport Reviews 42, 503–525. URL: https://www.sciencedirect.com/org/science/article/abs/pii/S0144164722004317, doi:10.1080/01441647.2021.2005183.

Lättman, K., Friman, M., Olsson, L.E., 2016. Perceived accessibility of public transport as a potential indicator of social inclusion. Social Inclusion 4, 36–45. URL: https://www.cogitatiopress.com/socialinclusion/article/view/481.

Martens, K., 2016a. Accessibility as primary good?. Routledge. Num Pages: 18.

Martens, K., 2016b. Defining sufficient accessibility. Routledge. Num Pages: 18.

Martens, K., Golub, A., 2021. A fair distribution of accessibility: Interpreting civil rights regulations for regional transportation plans. Journal of Planning Education and Research 41, 425–444. URL: https://doi.org/10.1177/0739456X18791014,

doi:10.1177/0739456X18791014.
Matas, A., Raymond, J.L., Roig, J.L., 2010. Job accessibility and female employment probability: The cases of barcelona and madrid. Urban Studies 47, 769–787. URL: https://doi.org/10.1177/0042098009352364, doi:10.1177/0042098009352364.
Mehdizadeh, M., Kroesen, M., Vos, J.D., 2025. Perceived accessibility and life satisfaction: the mediating role of activity participation? Transportation Research Part D: Transport and Environment 146, 104848. URL: https://www.sciencedirect.com/science/article/pii/S1361920925002585, doi:10.1016/j.trd.2025.104848.
Merlin, L.A., Hu, L., 2017. Does competition matter in measures of job accessibility? explaining employment in los angeles. Journal of Transport Geography 64, 77–88. URL: https://www.sciencedirect.com/science/article/pii/S0966692317302405, doi:10.1016/j.jtrangeo.2017.08.009.
Middleton, J., Spinney, J., 2019. Social inclusion, accessibility and emotional work. Policy Press. pp. 83–104. URL: https://www.degruyter.com/document/doi/10.56687/9781447329572-009/html?srsltid=AfmBOoqjdOQp3sJesUxgtJPEfg6LVN1eBYG9y6E9ev22CTBXZAruyRCd, doi:10.56687/9781447329572-009.
Miller, E.J., 2018. Accessibility: measurement and application in transportation planning. Transport Reviews 38, 551–555. URL: https://doi.org/10.1080/01441647.2018.1492778, doi:10.1080/01441647.2018.1492778. _eprint: https://doi.org/10.1080/01441647.2018.1492778.
Miller, H.J., 2005. Place-Based Versus People-Based Accessibility. Emerald Group Publishing Limited. pp. 63–89. URL: https://doi.org/10.1108/9780080460550-004, doi:10.1108/9780080460550-004.
Murphy, E.C., Porter, G., Aouidet, H., Dungey, C., Han, S., Houiji, R., Jlassi, M., Keskes, H., Mansour, H., Nasser, W., Riahi, H., Riahi, S., Zaghoud, H., 2023. 'no place for a woman': Access, exclusion, insecurity and the mobility regime in grand tunis. Geoforum 142, 103753. URL: https://www.sciencedirect.com/science/article/pii/S0016718523000799, doi:10.1016/j.geoforum.2023.103753.
Negm, H., De Vos, J., Pot, F., El-Geneidy, A., 2025. Perceived accessibility: A literature review. Journal of Transport Geography 125, 104212. URL: https://www.sciencedirect.com/science/article/pii/S0966692325001036, doi:10.1016/j.jtrangeo.2025.104212.
Negm, H., El-Geneidy, A., 2026. Operationalizing accessibility thresholds: Identifying perceived minimum and sufficient accessibility in canada. Journal of Transport Geography 135, 104758. URL: https://www.sciencedirect.com/science/article/pii/S0966692326002127, doi:10.1016/j.jtrangeo.2026.104758.
Owen, A., Levinson, D.M., 2015. Modeling the commute mode share of transit using continuous accessibility to jobs. Transportation Research Part A: Policy and Practice 74, 110–122. URL: https://www.sciencedirect.com/science/article/pii/S0965856415000191, doi:10.1016/j.tra.2015.02.002.
Paez, A., Mercado, R.G., Farber, S., Morency, C., Roorda, M., 2009. Mobility and social exclusion in canadian communities: An empirical investigation of opportunity access and deprivation from the perspective of vulnerable groups .
Palm, M., Teel, S., Tiznado-Aitken, I., Soukhov, A., Paez, A., Farber, S., Hain, M., 2023. Developing Data Driven Equity Standards: Stakeholder Perspectives. Technical Report. URL: https://mobilizingjustice.ca/wp-content/uploads/2023/06/Developing_Data_Driven_Standards_FINAL_2023_06_06.pdf.
Pebesma, E., 2018. Simple Features for R: Standardized Support for Spatial Vector Data. The R Journal 10, 439–446. URL: https://doi.org/10.32614/RJ-2018-009, doi:10.32614/RJ-2018-009.
Pereira, R.H.M., Herszenhut, D., 2023. What is accessibility?. Rio de Janeiro. URL: https://ipeagit.github.io/intro_access_book, doi:10.38116/9786556350653.
Pereira, R.H.M., Karner, A., 2021. Transportation Equity. Elsevier, Oxford. pp. 271–277. URL: https://www.sciencedirect.com/science/article/pii/B9780081026717100533, doi:10.1016/B978-0-08-102671-7.10053-3.
Pereira, R.H.M., Saraiva, M., Herszenhut, D., Braga, C.K.V., Conway, M.W., 2021. r5r: Rapid realistic routing on multimodal transport networks with r^5 in r. Findings URL: https://findingspress.org/article/21262-r5r-rapid-realistic-routing-on-multimodal-transport-networks-with-r-5-in-r, doi:10.32866/001c.21262.
Pereira, R.H.M., Schwanen, T., Banister, D., 2017. Distributive justice and equity in transportation. Transport Reviews 37, 170–191. URL: https://doi.org/10.1080/01441647.2016.1257660, doi:10.1080/01441647.2016.1257660. _eprint: https://doi.org/10.1080/01441647.2016.1257660.
Peungnumsai, A., Miyazaki, H., Witayangkurn, A., Kim, S.M., 2020. A grid-based spatial analysis for detecting supply–demand gaps of public transports: A case study of the bangkok metropolitan region. Sustainability 12, 10382. URL: https://www.mdpi.com/2071-1050/12/24/10382, doi:10.3390/su122410382. number: 24.
Pot, F.J., 2025. Perceived accessibility and transport equity: Does satisfaction imply perceived sufficiency of opportunities? Transportation Research Part A: Policy and Practice 198, 104497. URL: https://www.sciencedirect.com/science/article/pii/S0965856425001259, doi:10.1016/j.tra.2025.104497.
Pot, F.J., Van Wee, B., Tillema, T., 2021. Perceived accessibility: What it is and why it differs from calculated accessibility measures based on spatial data. Journal of Transport Geography 94, 103090. URL: https://linkinghub.elsevier.com/retrieve/pii/S0966692321001435, doi:10.1016/j.jtrangeo.2021.103090.
Preston, J., Rajé, F., 2007. Accessibility, mobility and transport-related social exclusion. Journal of Transport Geography 15, 151–160. URL: https://www.sciencedirect.com/science/article/pii/S0966692306000512, doi:10.1016/j.jtrangeo.2006.05.002.
Páez, A., 2021. Open spatial sciences: an introduction. Journal of Geographical Systems 23, 467–476. URL: https://doi.org/10.1007/s10109-021-00364-4, doi:10.1007/s10109-021-00364-4.
Páez, A., Scott, D.M., Morency, C., 2012. Measuring accessibility: positive and normative implementations of various accessibility indicators. Journal of Transport Geography 25, 141–153. URL: https://www.sciencedirect.com/science/article/pii/S0966692312000798, doi:10.1016/j.jtrangeo.2012.03.016.
Rawls, J., 1999. A Theory of Justice. volume 1. Harvard University Press, Cambridge, MA. URL: https://www.jstor.org/

stable/j.ctvjf9z6v, doi:10.2307/j.ctvjf9z6v.
dos Santos, B.D., Moghadasi, M., Páez, A., 2026. A 30-year analysis of active travel behaviour in canada. Journal of Transport Geography 131, 104561. URL: https://www.sciencedirect.com/science/article/pii/S0966692326000153, doi:10.1016/j.jtrangeo.2026.104561.
Sen, A., 1979. Equality of what? the tanner lecture on human values. Stanford University 22.
Sen, A., 1999. Development as freedom. Anchor, New York.
Sen, A., 2005. Human rights and capabilities. Journal of Human Development 6, 151–166. URL: https://doi.org/10.1080/14649880500120491, doi:10.1080/14649880500120491. _eprint: https://doi.org/10.1080/14649880500120491.
Sen, A., 2009. The Idea of Justice. Harvard University Press. URL: https://www.jstor.org/stable/j.ctvjnrv7n, doi:10.2307/j.ctvjnrv7n.
Shen, Q., 1998. Location characteristics of inner-city neighborhoods and employment accessibility of low-wage workers. Environment and Planning B: Planning and Design 25, 345–365. URL: https://journals.sagepub.com/doi/abs/10.1068/b250345, doi:10.1068/b250345.
Sjoberg, D.D., Whiting, K., Curry, M., Lavery, J.A., Larmarange, J., 2021. Reproducible summary tables with the gtsummary package. The R Journal 13, 570–580. URL: https://doi.org/10.32614/RJ-2021-053, doi:10.32614/RJ-2021-053.
Soukhov, A., Paez, A., 2024. Accessibility analysis for planning applications. Technical Report. URL: https://github.com/soukhova/MJ-Accessibility-Blogs.
Soukhov, A., Pereira, R.H.M., Higgins, C.D., Páez, A., 2025a. A family of accessibility measures derived from spatial interaction principles. PLOS ONE 20, e0335951. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0335951, doi:10.1371/journal.pone.0335951.
Soukhov, A., Páez, A., Higgins, C.D., Mohamed, M., 2023. Introducing spatial availability, a singly-constrained measure of competitive accessibility. PLOS ONE 18, e0278468. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0278468, doi:10.1371/journal.pone.0278468.
Soukhov, A., Tarriño-Ortiz, J., Soria-Lara, J.A., Páez, A., 2024. Multimodal spatial availability: A singly-constrained measure of accessibility considering multiple modes. PLOS ONE 19, e0299077. URL: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0299077, doi:10.1371/journal.pone.0299077.
Soukhov, A., Tiznado-Aitken, I., Palm, M., Farber, S., Paez, A., 2025b. Standards of fairness in the transportation literature. Transportation URL: https://doi.org/10.1007/s11116-025-10693-8, doi:10.1007/s11116-025-10693-8.
Sun, Y., Thakuriah, P., 2021. Public transport availability inequalities and transport poverty risk across england. Environment and Planning B: Urban Analytics and City Science 48, 2775–2789. URL: https://doi.org/10.1177/2399808321991536, doi:10.1177/2399808321991536.
Tabascio, A., Tiznado-Aitken, I., Higgins, C., Farber, S., 2024. Incorporating equity into transit performance measures: A disaggregated bus route level approach. Case Studies on Transport Policy 17, 101256. URL: https://www.sciencedirect.com/science/article/pii/S2213624X24001111, doi:10.1016/j.cstp.2024.101256.
te Brömmelstroet, M., Curtis, C., Larsson, A., Milakis, D., 2016. Strengths and weaknesses of accessibility instruments in planning practice: technological rules based on experiential workshops. European Planning Studies 24, 1175–1196. URL: https://doi.org/10.1080/09654313.2015.1135231, doi:10.1080/09654313.2015.1135231. _eprint: https://doi.org/10.1080/09654313.2015.1135231.
Tennekes, M., 2018. tmap: Thematic maps in r. Journal of Statistical Software 84, 1–39. URL: https://www.jstatsoft.org/index.php/jss/article/view/v084i06, doi:10.18637/jss.v084.i06.
Terza, J.V., Basu, A., Rathouz, P.J., 2008. Two-stage residual inclusion estimation: Addressing endogeneity in health econometric modeling. Journal of Health Economics 27, 531–543. URL: https://www.sciencedirect.com/science/article/pii/S0167629607001063, doi:10.1016/j.jhealeco.2007.09.009.
Tiznado-Aitken, I., Muñoz, J.C., Hurtubia, R., 2018. The role of accessibility to public transport and quality of walking environment on urban equity: The case of santiago de chile. Transportation Research Record 2672, 129–138. URL: https://doi.org/10.1177/0361198118782036, doi:10.1177/0361198118782036.
Tiznado-Aitken, I., Singer, M., Morency, C., Hassan, H., Morissette, S.D., Verreault, H., Palm, M., Farber, S., 2025. The first national survey on transport poverty: Design, implementation and findings from canada. OSF Preprints doi:10.31235/osf.io/j2pkm_v1.
Upham, P., Sovacool, B.K., Monyei, C.G., 2022. Energy and transport poverty amidst plenty: Exploring just transition, lived experiences and policy implications in iceland. Renewable and Sustainable Energy Reviews 163, 112533. URL: https://www.sciencedirect.com/science/article/pii/S1364032122004348, doi:10.1016/j.rser.2022.112533.
van Dülmen, C., Šimon, M., Klärner, A., 2022. Transport poverty meets car dependency: A gps tracking study of socially disadvantaged groups in european rural peripheries. Journal of Transport Geography 101, 103351. URL: https://www.sciencedirect.com/science/article/pii/S0966692322000746, doi:10.1016/j.jtrangeo.2022.103351.
Verma, R., Ukkusuri, S.V., 2025. What determines travel time and distance decay in spatial interaction and accessibility? Journal of Transport Geography 122, 104061. URL: https://www.sciencedirect.com/science/article/pii/S0966692324002709, doi:10.1016/j.jtrangeo.2024.104061.
van der Vlugt, A.L., Curl, A., Scheiner, J., 2022. The influence of travel attitudes on perceived walking accessibility and walking behaviour. Travel Behaviour and Society 27, 47–56. URL: https://www.sciencedirect.com/science/article/pii/S2214367X2100106X, doi:10.1016/j.tbs.2021.11.002.
von Bergmann, J., Shkolnik, D., Jacobs, A., 2021. cancensus: R package to access, retrieve, and work with Canadian Census data and geography. URL: https://mountainmath.github.io/cancensus/. r package version 0.4.2.
Wachs, M., Kumagai, T.G., 1973. Physical accessibility as a social indicator. Socio-Economic Planning Sciences 7, 437–456. URL: https://www.sciencedirect.com/science/article/pii/0038012173900414, doi:10.1016/0038-0121(73)90041-4.

Wee, B.v., Geurs, K., 2011. Discussing equity and social exclusion in accessibility evaluations. European Journal of Transport and Infrastructure Research 11. URL: https://journals.open.tudelft.nl/ejtir/article/view/2940, doi:10.18757/ejtir.2011.11.4.2940. number: 4.

Wickham, H., 2016. ggplot2: Elegant Graphics for Data Analysis. Springer-Verlag New York. URL: https://ggplot2.tidyverse.org.

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L.D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T.L., Miller, E., Bache, S.M., Müller, K., Ooms, J., Robinson, D., Seidel, D.P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., Yutani, H., 2019. Welcome to the tidyverse. Journal of Open Source Software 4, 1686. doi:10.21105/joss.01686.

Wickham, H., François, R., Henry, L., Müller, K., Vaughan, D., 2026. dplyr: A Grammar of Data Manipulation. URL: https://CRAN.R-project.org/package=dplyr, doi:10.32614/CRAN.package.dplyr. r package version 1.2.1.

Wickham, H., Vaughan, D., Girlich, M., 2025. tidyr: Tidy Messy Data. URL: https://CRAN.R-project.org/package=tidyr, doi:10.32614/CRAN.package.tidyr. r package version 1.3.2.

Wooldridge, J.M., 2010. Econometric analysis of cross section and panel data. MIT press.

Yenisetty, P.T., Bahadure, P., 2020. Measuring accessibility to various asfs from public transit using spatial distance measures in indian cities. ISPRS International Journal of Geo-Information 9, 446. URL: https://www.mdpi.com/2220-9964/9/7/446, doi:10.3390/ijgi9070446. number: 7.

Zhu, L., Lucas, K., 2025. Towards calculated and perceived transport equity: An equity evaluation framework for accessibility. Transportation Research Part D: Transport and Environment 146, 104908. URL: https://www.sciencedirect.com/science/article/pii/S1361920925003189, doi:10.1016/j.trd.2025.104908.